KỸ THUẬT SỐ

Đánh giá

- **Điểm quá trình: 50%**
  - Chuyên cần
  - Kiểm tra
  - Tiểu luận
- **Điểm cuối kỳ: 50%**

Tài liệu tham khảo

- Nguyễn Trường Duy, Giáo trình Kỹ thuật số, Đại học Quốc gia Tp. HCM, Năm 2019.
- Ronald J. Tocci - Neal S. Widmer – Gregory L. Moss, Digital Systems: Principles and Applications, 12th Ed. Prentice Hall, 2017.
- Thomas L. Floyd, Digital Fundamental, 11th Ed. Prentice Hall, 2015.

Đại số Boolean – Cổng Logic

Nội dung

- Đại số Boolean
- Hàm Logic
- Bìa Karnaugh
- Mạch logic

Đại số Boolean

- Các thuật ngữ

Logic 0	Logic 1
False	True
Off	On
LOW	HIGH
No	Yes
Open switch	Closed switch

Đại số Boolean

- Bảng sự thật (trạng thái)

Output

Inputs

A	B	x
0	0	1
0	1	0
1	0	1
1	1	0

(a)

A	B	C	x
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1

(b)

A	B	C	D	x
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	0
1	0	0	1	0
1	0	1	0	0
1	0	1	1	1
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1

(c)

Đại số Boolean

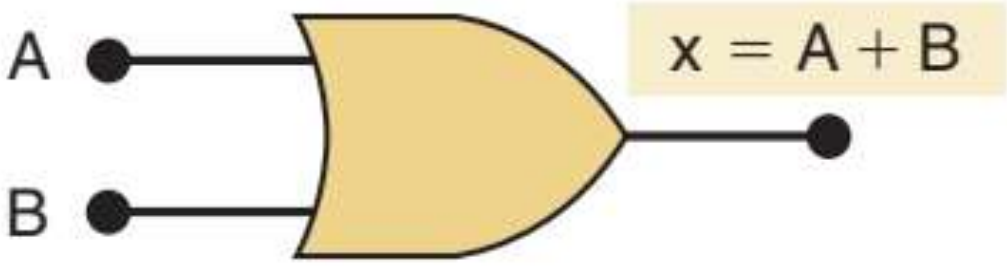
- Toán tử OR và cổng OR

$$x = A + B$$

OR

A	B	x = A + B
0	0	0
0	1	1
1	0	1
1	1	1

(a)



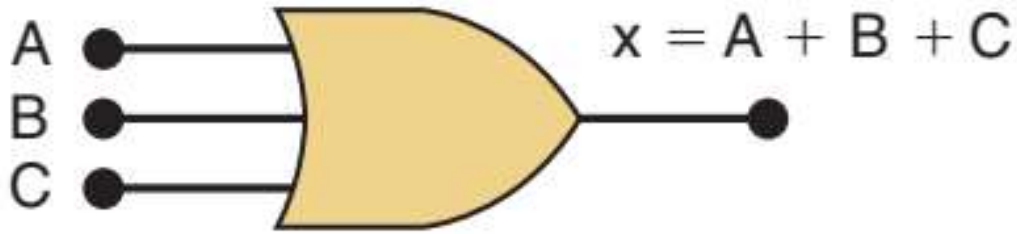
OR Gate

(b)

Đại số Boolean

- Toán tử OR và cổng OR

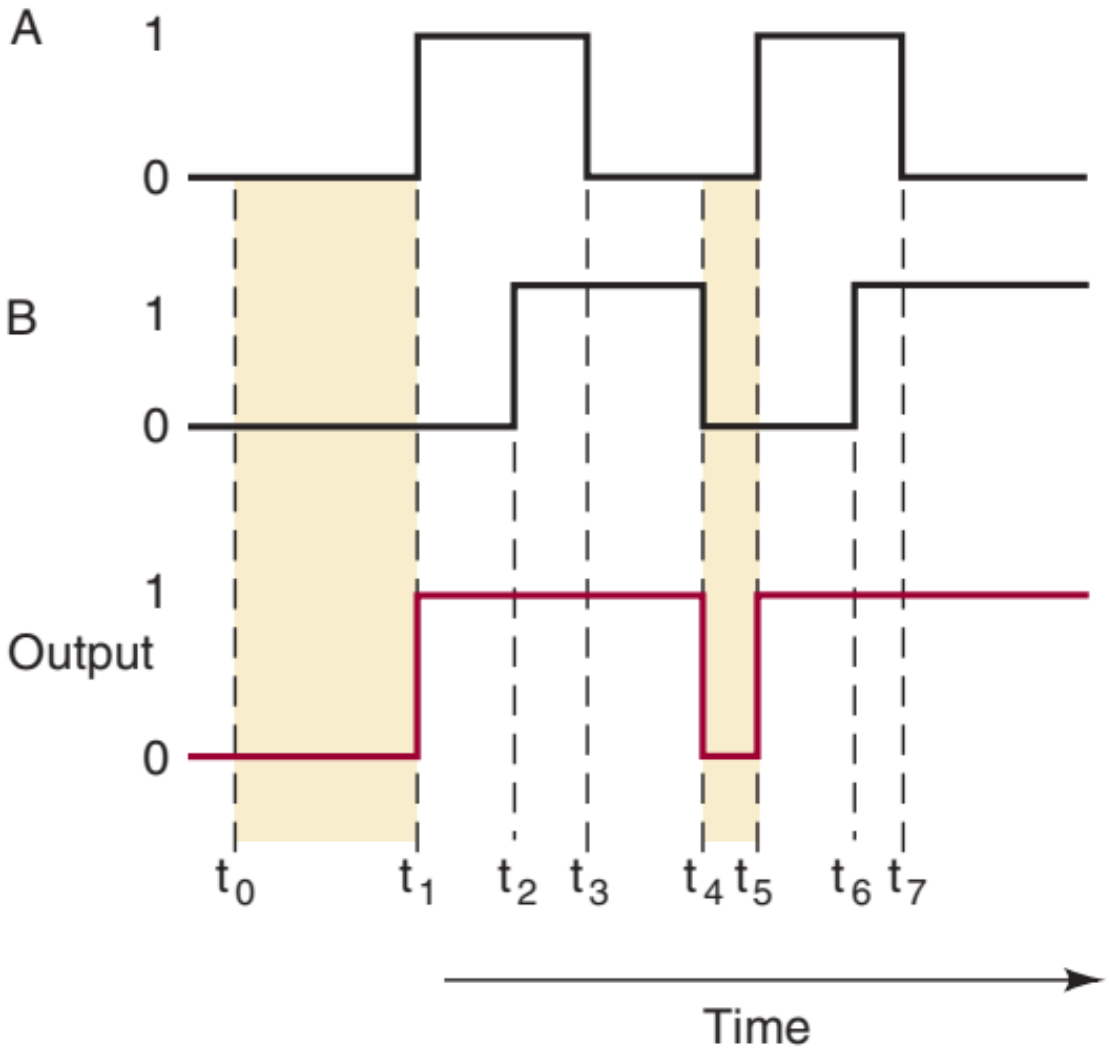
$$x = A + B + C$$

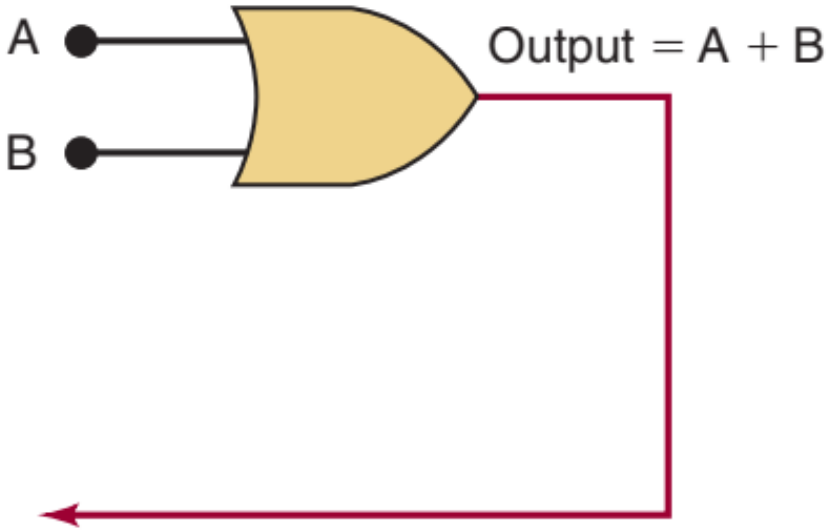


A	B	C	x = A + B + C
0	0	0	0
0	0	1	1
0	1	0	1
0	1	1	1
1	0	0	1
1	0	1	1
1	1	0	1
1	1	1	1

Đại số Boolean

- Toán tử OR và cổng OR





Đại số Boolean

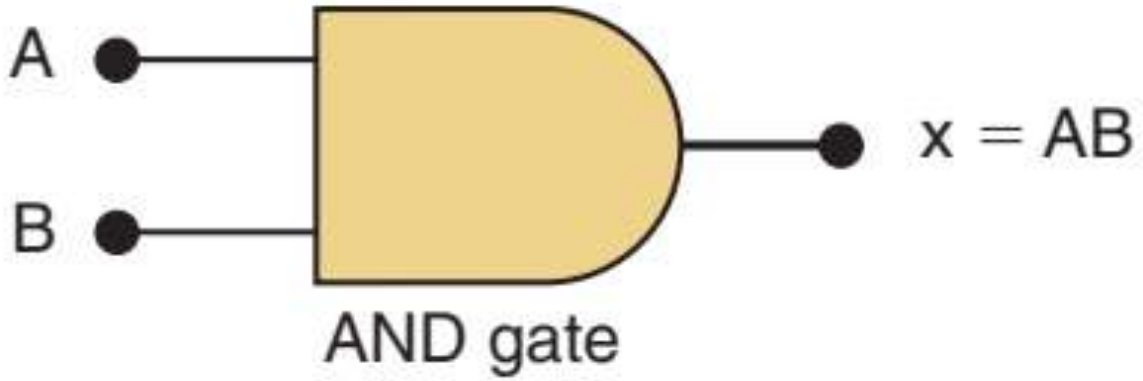
- Toán tử AND và cổng AND

$$x = A \cdot B = AB$$

AND

A	B	$x = A \cdot B$
0	0	0
0	1	0
1	0	0
1	1	1

(a)



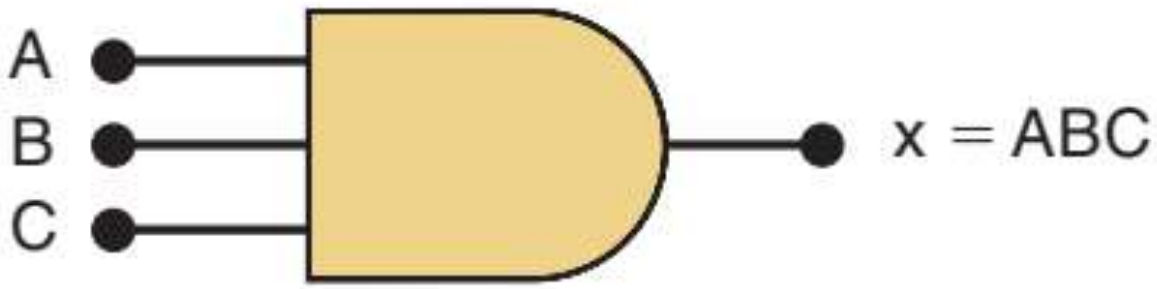
(b)

Đại số Boolean

- Toán tử AND và cổng AND

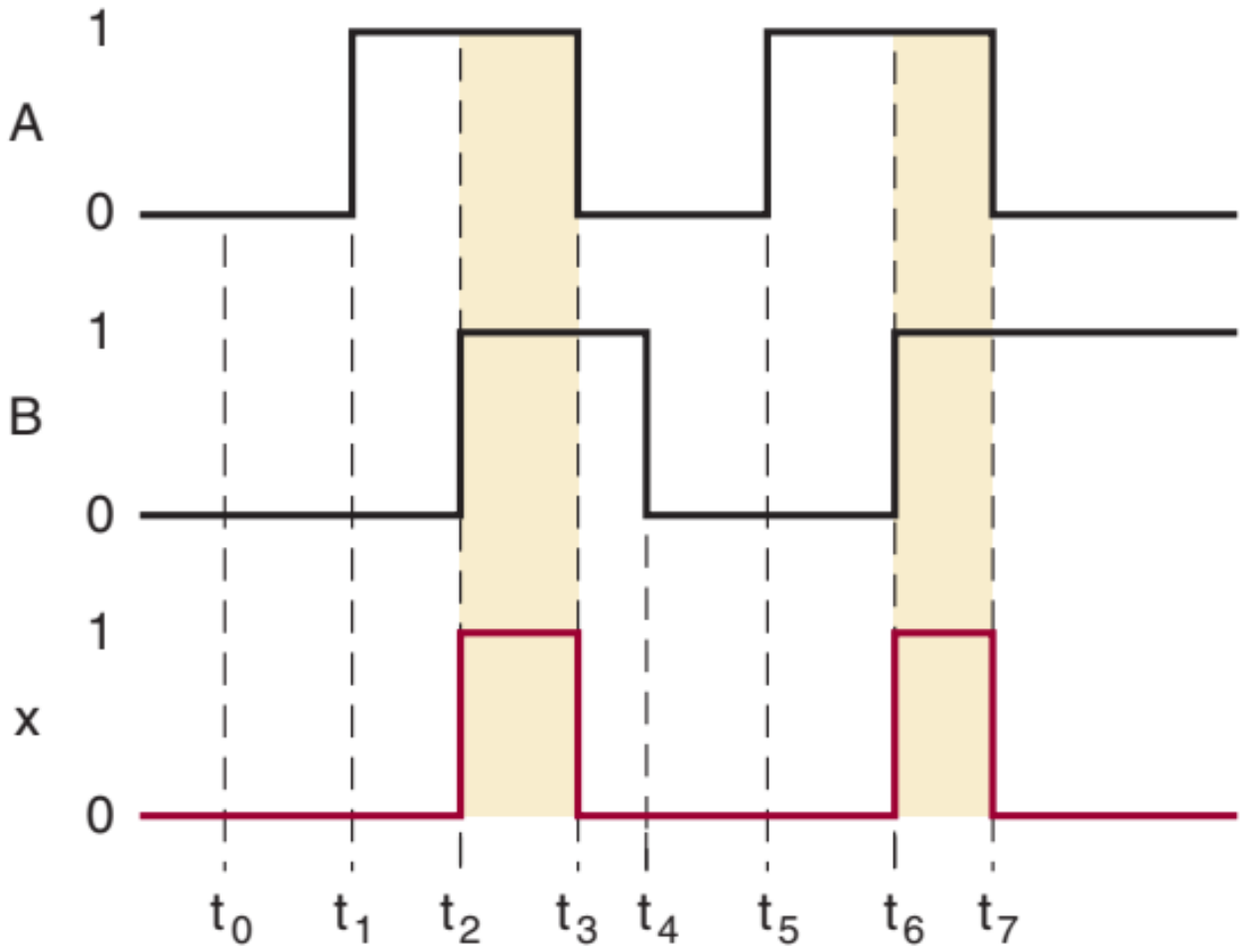
$$x = A \cdot B \cdot C = ABC.$$

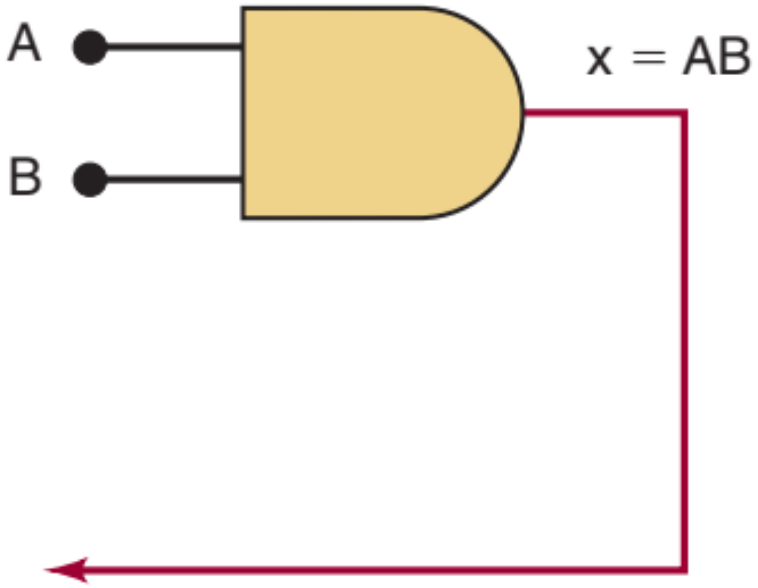
A	B	C	x = ABC
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	0
1	0	1	0
1	1	0	0
1	1	1	1



Đại số Boolean

- Toán tử AND và cổng AND





Đại số Boolean

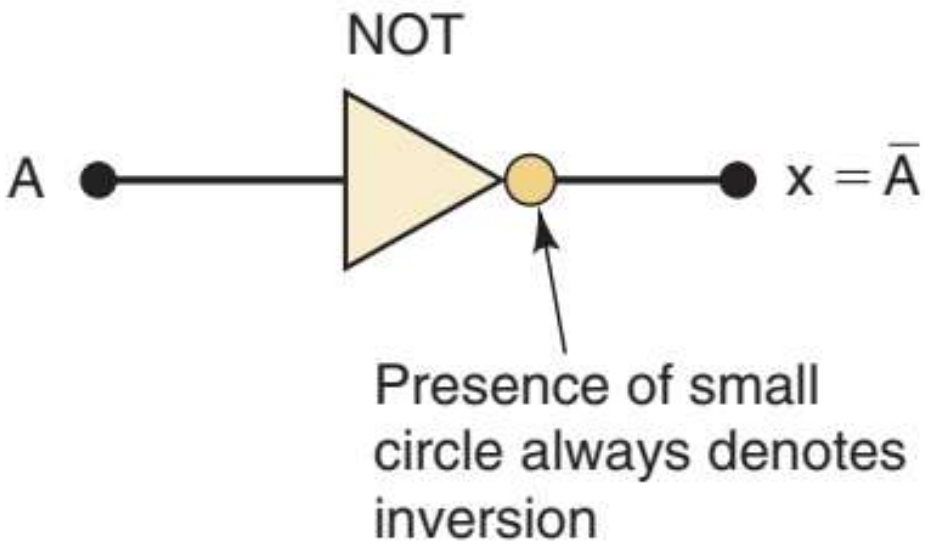
- Toán tử NOT và cổng NOT

$$x = \overline{A}$$

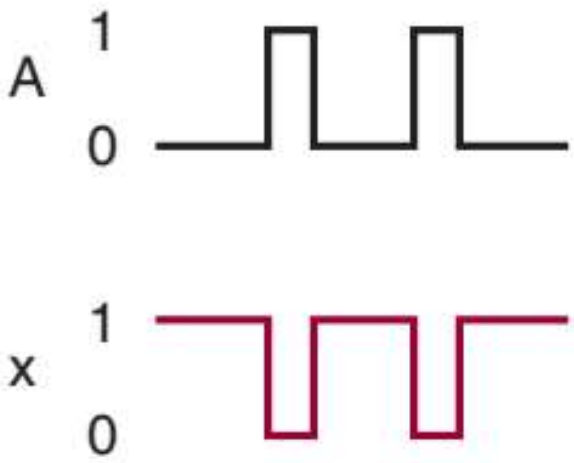
NOT

A	$x = \overline{A}$
0	1
1	0

(a)



(b)



(c)

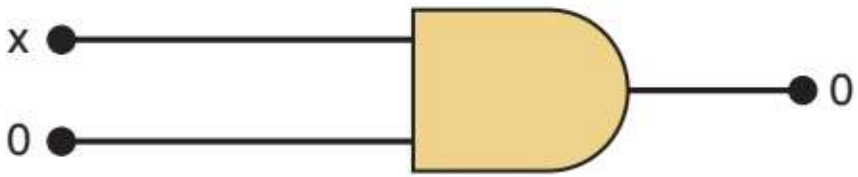
Đại số Boolean

- Các toán tử

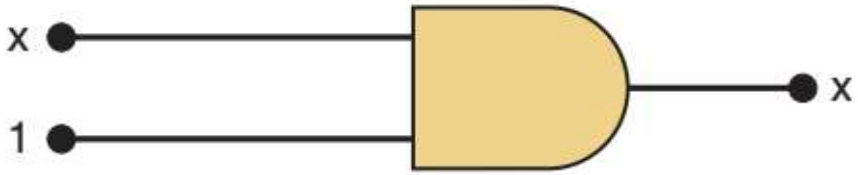
OR	AND	NOT
$0 + 0 = 0$	$0 \cdot 0 = 0$	$\overline{0} = 1$
$0 + 1 = 1$	$0 \cdot 1 = 0$	$\overline{1} = 0$
$1 + 0 = 1$	$1 \cdot 0 = 0$	
$1 + 1 = 1$	$1 \cdot 1 = 1$	

Đại số Boolean

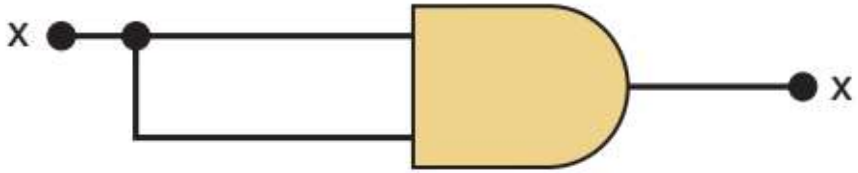
- Các định lý



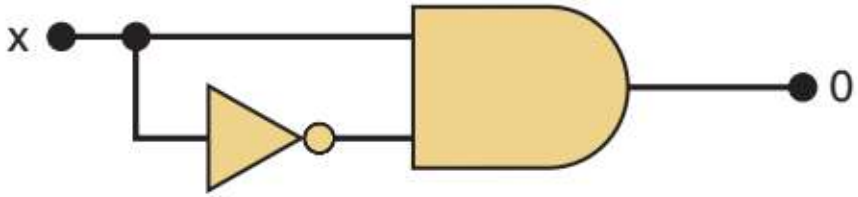
(1) $x \cdot 0 = 0$



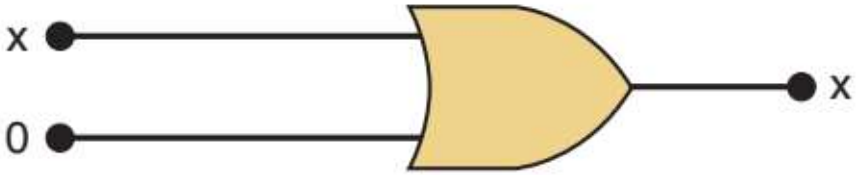
(2) $x \cdot 1 = x$



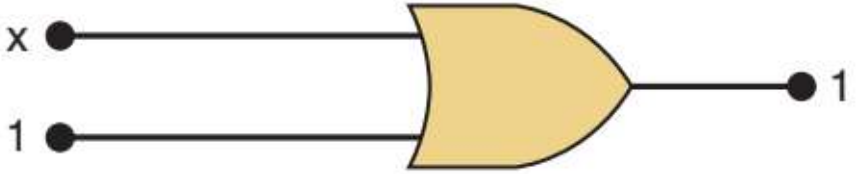
(3) $x \cdot x = x$



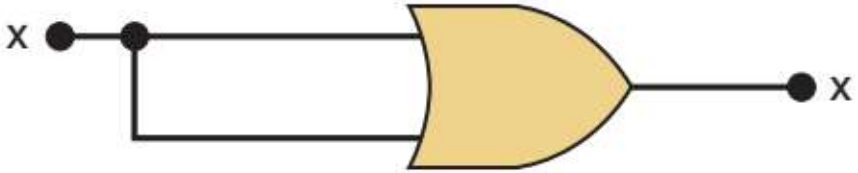
(4) $x \cdot \bar{x} = 0$



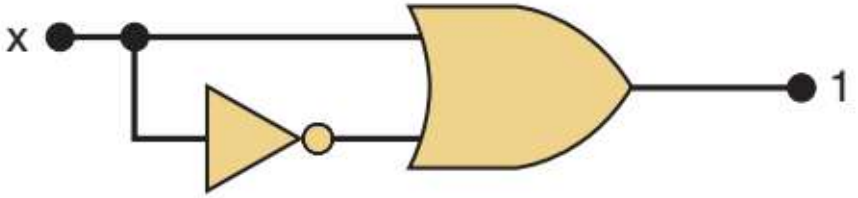
(5) $x + 0 = x$



(6) $x + 1 = 1$



(7) $x + x = x$



(8) $x + \bar{x} = 1$

Đại số Boolean

- Các định lý

(9) $x + y = y + x$

(10) $x \cdot y = y \cdot x$

(11) $x + (y + z) = (x + y) + z = x + y + z$

(12) $x(yz) = (xy)z = xyz$

(13a) $x(y + z) = xy + xz$

(13b) $(w + x)(y + z) = wy + xy + wz + xz$

(14) $x + xy = x$

(15a) $x + \bar{x}y = x + y$

(15b) $\bar{x} + xy = \bar{x} + y$

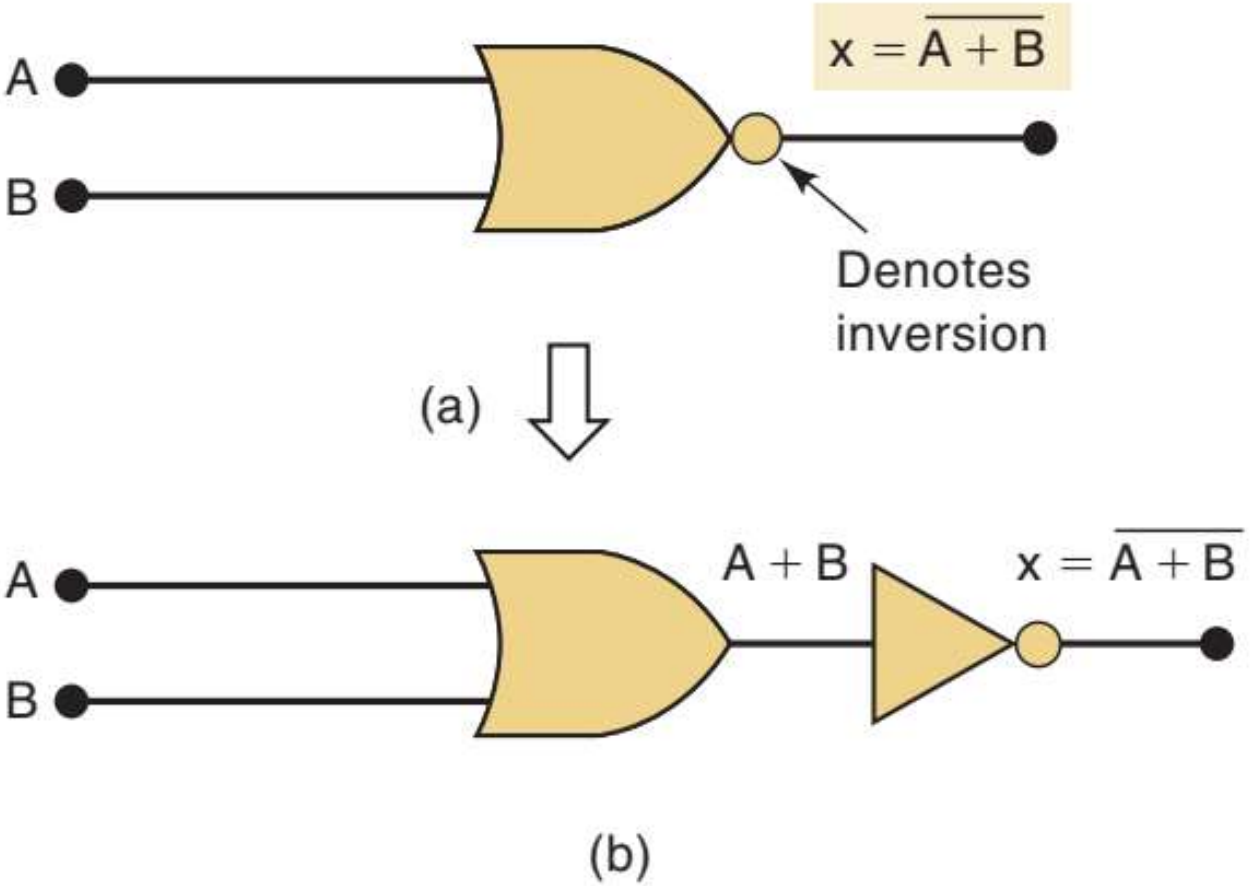
(16) $\overline{(x + y)} = \bar{x} \cdot \bar{y}$

(17) $\overline{(x \cdot y)} = \bar{x} + \bar{y}$

Đại số Boolean

- Cổng NOR

$$x = \overline{A + B}$$

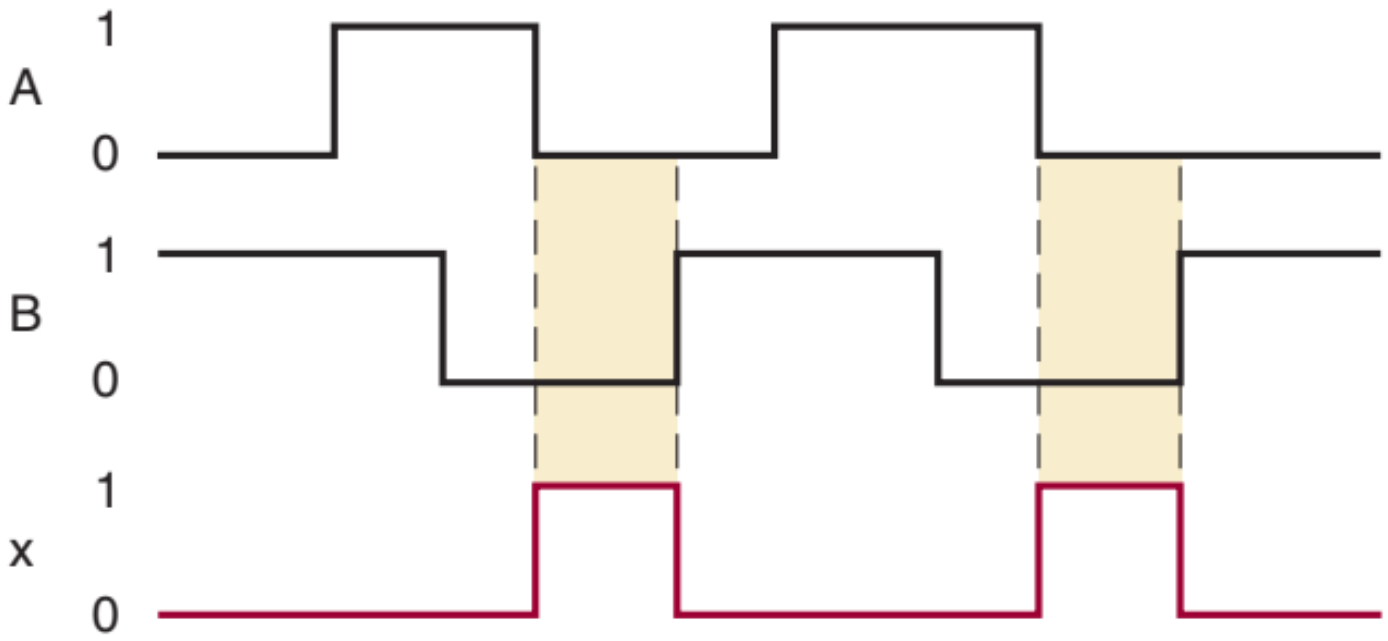


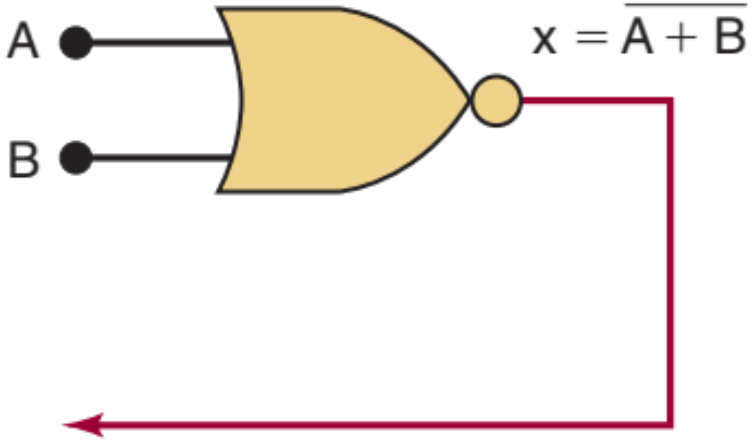
		OR	NOR
A	B	$A + B$	$\overline{A + B}$
0	0	0	1
0	1	1	0
1	0	1	0
1	1	1	0

(c)

Đại số Boolean

- Cổng NOR

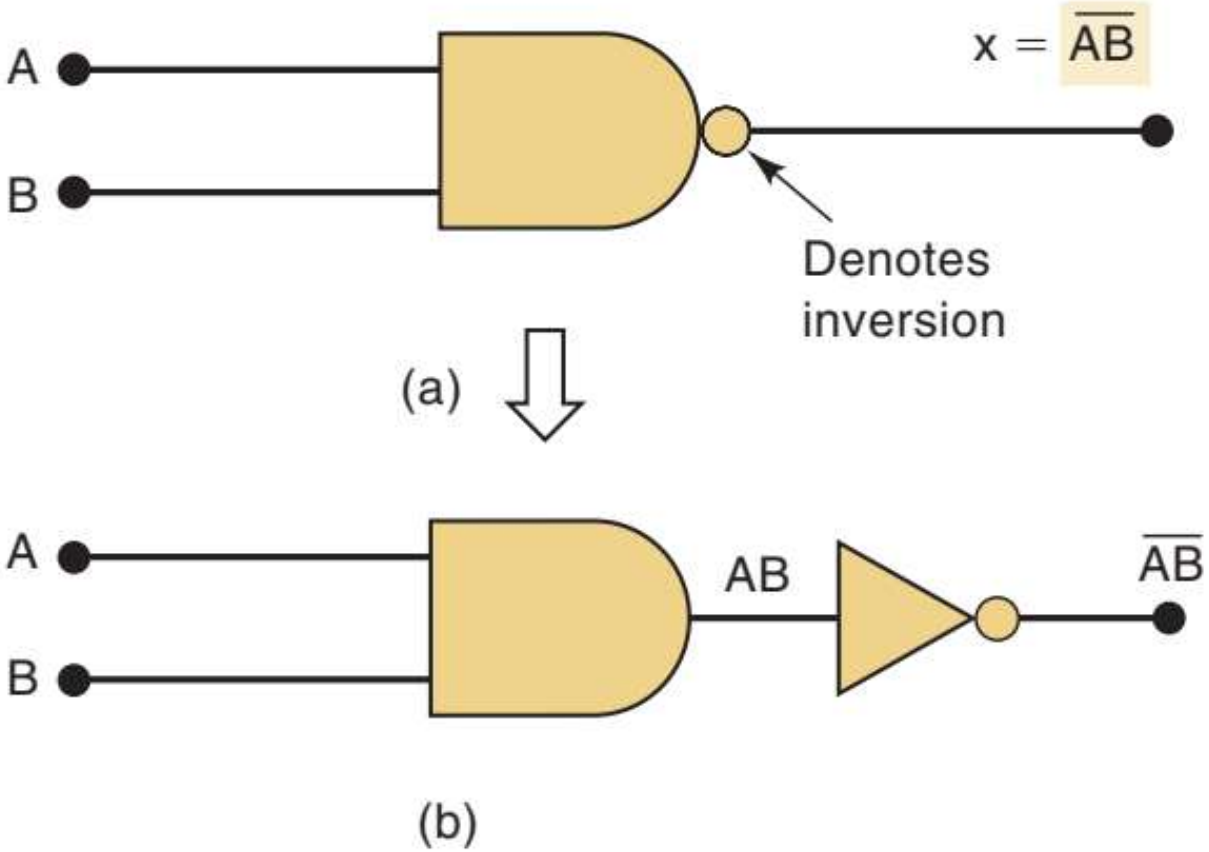




Đại số Boolean

- Cổng NAND

$$x = \overline{AB}$$

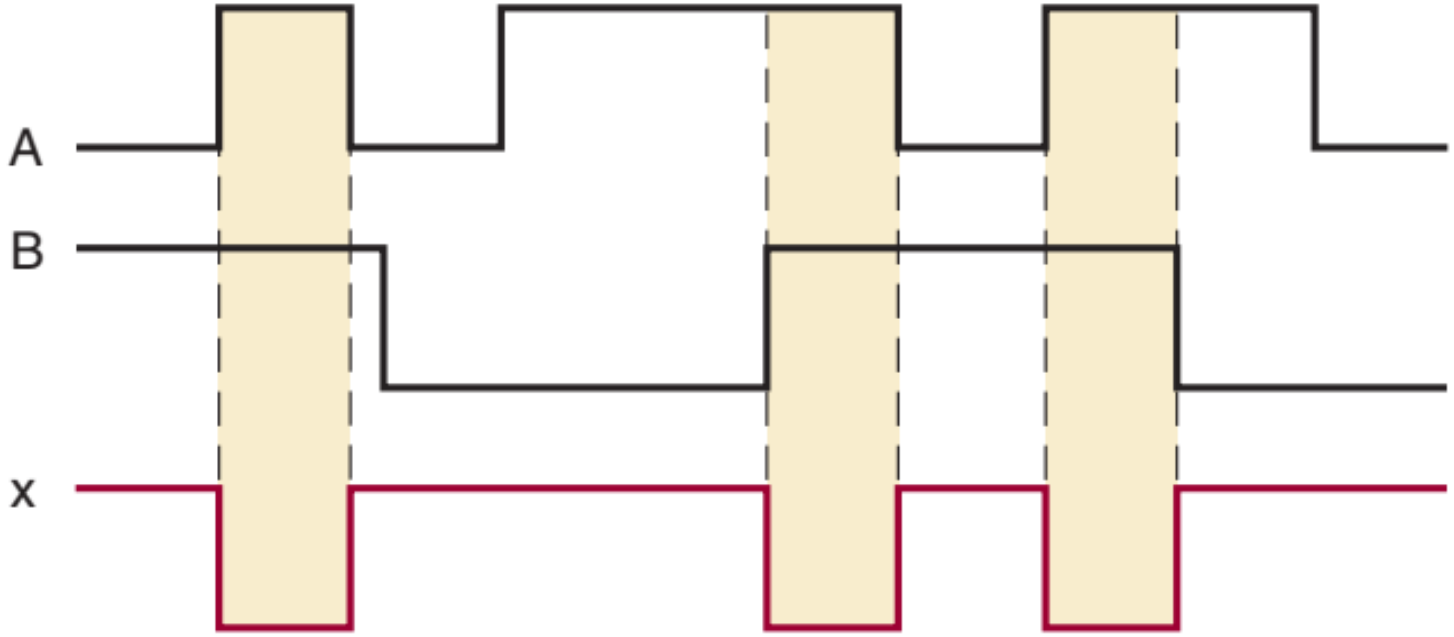


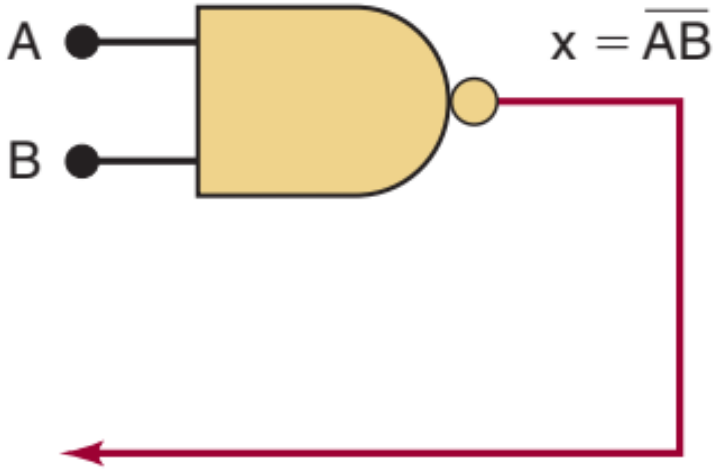
		AND	NAND
A	B	AB	$\overline{AB}$
0	0	0	1
0	1	0	1
1	0	0	1
1	1	1	0

(c)

Đại số Boolean

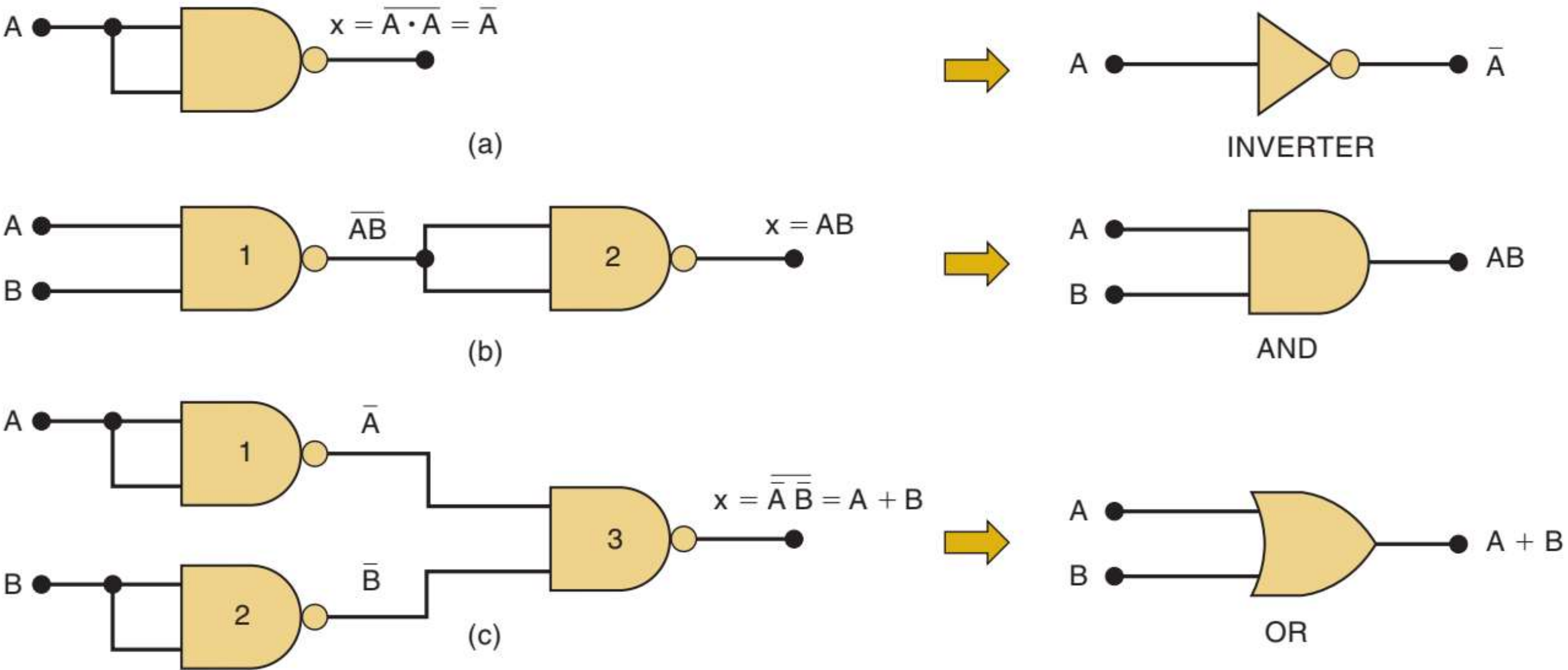
- Cổng NAND





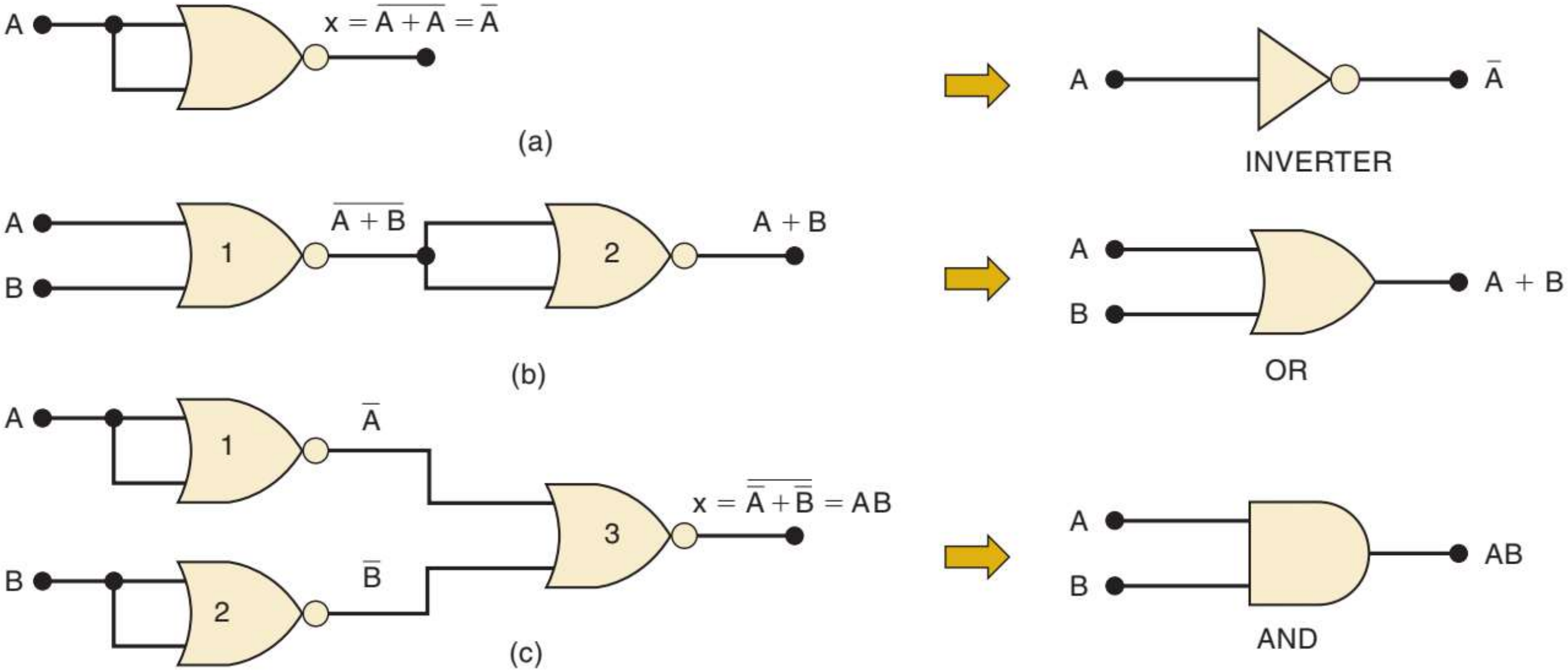
Đại số Boolean

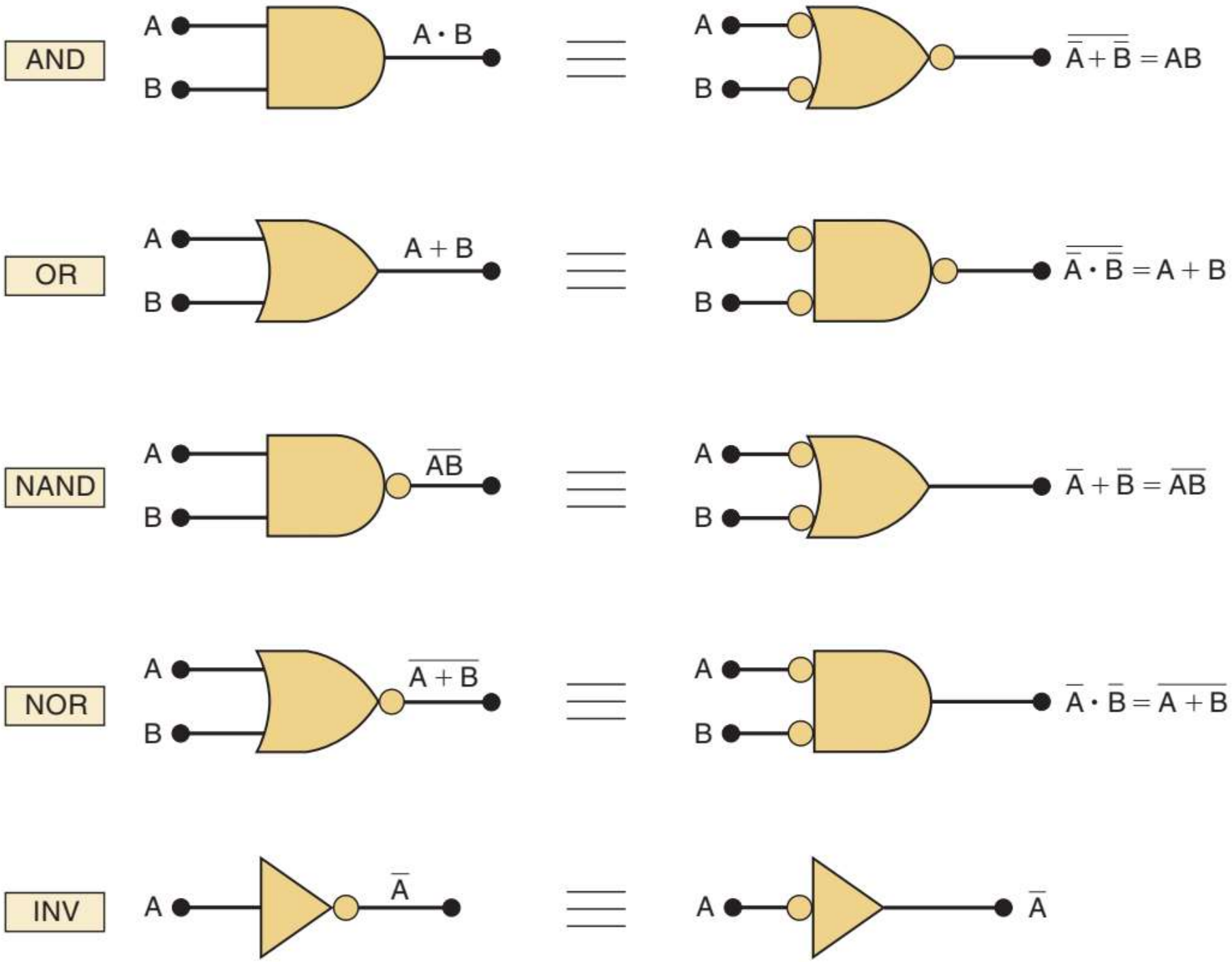
- Cổng đa năng NAND



Đại số Boolean

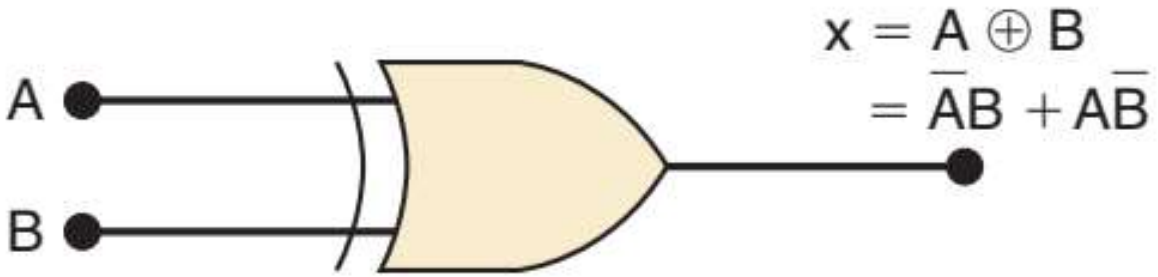
- Cổng đa năng NOR



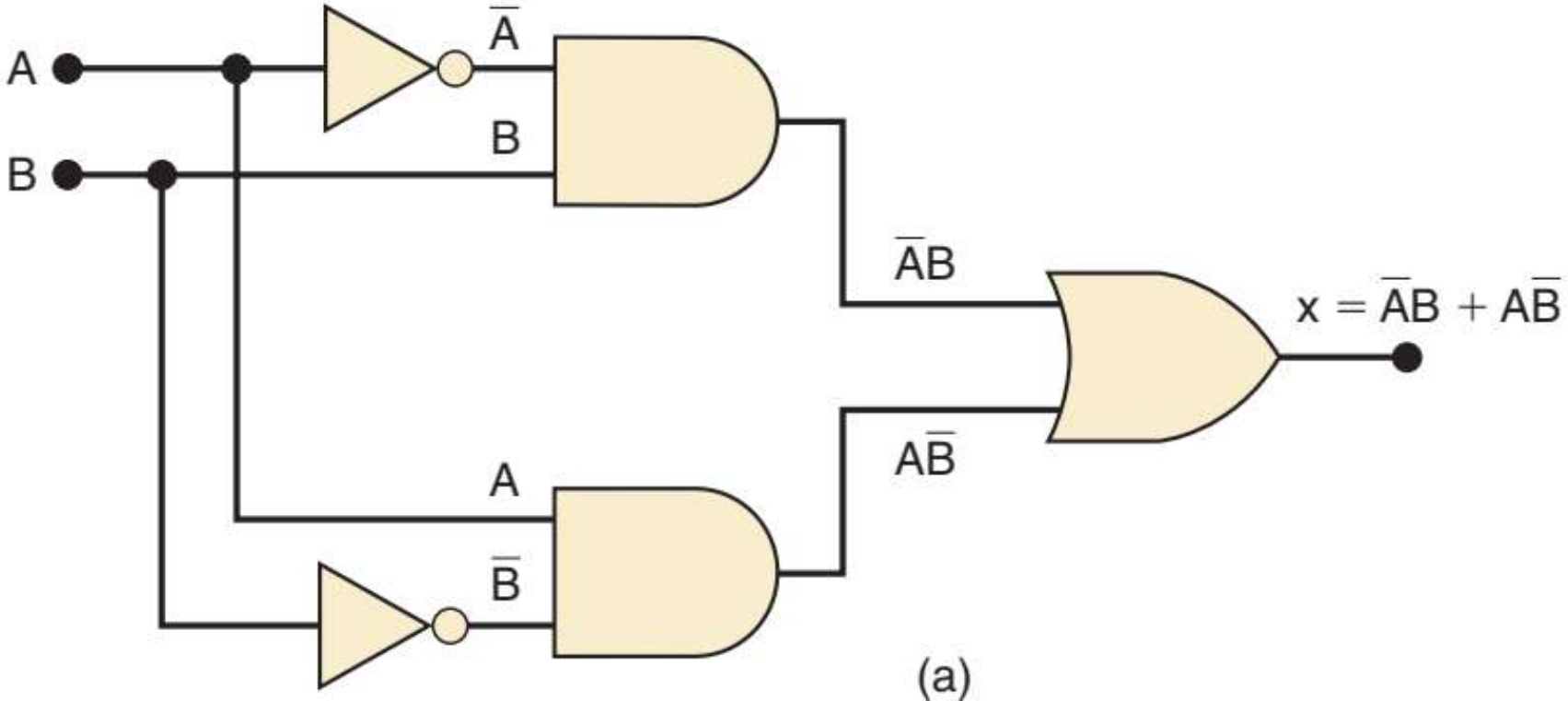

AND
A
B
A · B
A
B
$\overline{\bar{A}+\bar{B}} = AB$
OR
A
B
A + B
A
B
$\overline{\bar{A}\cdot\bar{B}} = A + B$
NAND
A
B
$\overline{AB}$
A
B
$\bar{A}+\bar{B} = \overline{AB}$
NOR
A
B
$\overline{A+B}$
A
B
$\bar{A}\cdot\bar{B} = \overline{A+B}$
INV
A
$\bar{A}$
A
$\bar{A}$

Đại số Boolean

- Cổng EX-OR (XOR)



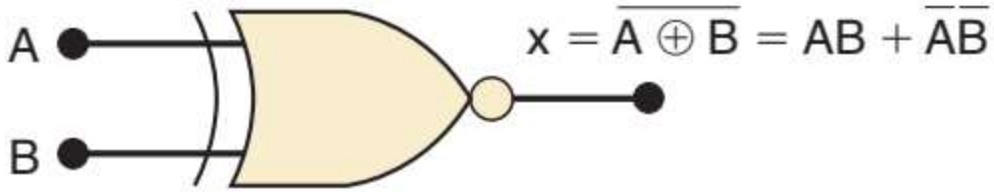
$$x = \bar{A}B + A\bar{B}$$



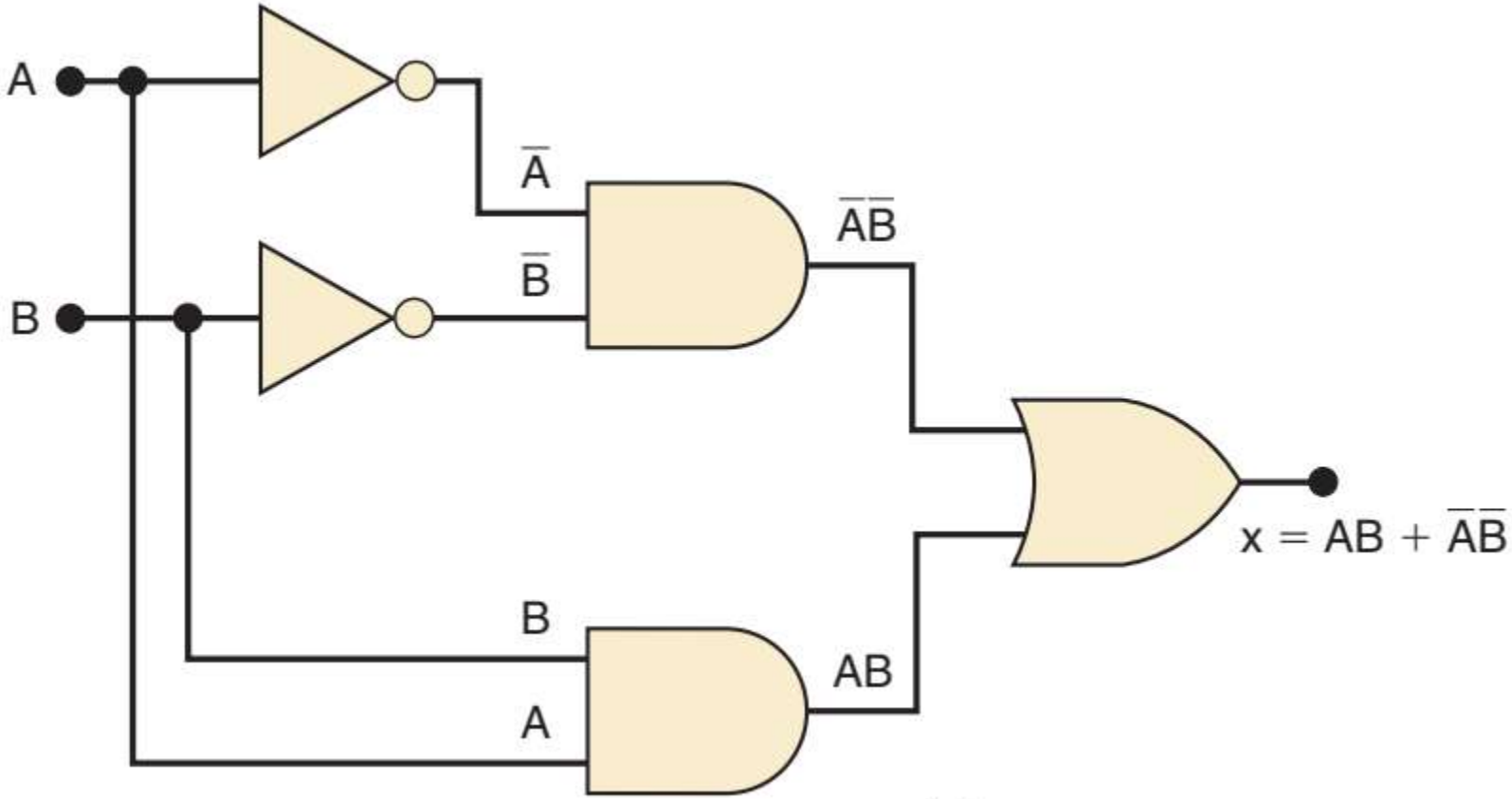
A	B	x
0	0	0
0	1	1
1	0	1
1	1	0

Đại số Boolean

- Cổng EX-NOR (XNOR)



$$x = AB + \overline{A}\,\overline{B}$$



A	B	x
0	0	1
0	1	0
1	0	0
1	1	1

Hàm Logic

- Một số hàm Logic

$$F = (x \,.\, y) + z'$$

$$F_1 = (x \,.\, z) + y'$$

$$F_2 = (x \,.\, y \,.\, z) + (y \,.\, z')$$

$$F(x, y, z) = (x \,.\, y) + z'$$

$$F_1(x, y, z) = (x \,.\, z) + y'$$

$$F_2(x, y, z) = (x \,.\, y \,.\, z) + (y \,.\, z')$$

Hàm Logic

- Một số hàm Logic

a) $F_1 = xy + xy' + y'z$

Table 1: Truth Table of Function, F_1

Input Variables and Complement				Product Terms			Function Output
x	y	z	y'	xy	xy'	$y'z$	F_1
0	0	0	1	0	0	0	0
0	0	1	1	0	0	1	1
0	1	0	0	0	0	0	0
0	1	1	0	0	0	0	0
1	0	0	1	0	1	0	1
1	0	1	1	0	1	1	1
1	1	0	0	1	0	0	1
1	1	1	0	1	0	0	1

Hàm Logic

- Minterns

x	y	Minterm	Notation
0	0	$x'.y'$	m_0
0	1	$x'.y$	m_1
1	0	$x.y'$	m_2
1	1	$x.y$	m_3

Hàm Logic

- Minterns

x	y	z	Minterm	Notation
0	0	0	$x'.y'.z'$	m_0
0	0	1	$x'.y'.z$	m_1
0	1	0	$x'.y.z'$	m_2
0	1	1	$x'.y.z$	m_3
1	0	0	$x.y'.z'$	m_4
1	0	1	$x.y'.z$	m_5
1	1	0	$x.y.z'$	m_6
1	1	1	$x.y.z$	m_7

Hàm Lo

- Minterns

a	b	c	d	Minterm	Notation
0	0	0	0	$a'.b'.c'.d'$	m_0
0	0	0	1	$a'.b'.c'.d$	m_1
0	0	1	0	$a'.b'.c.d'$	m_2
0	0	1	1	$a'.b'.c.d$	m_3
0	1	0	0	$a'.b.c'.d'$	m_4
0	1	0	1	$a'.b.c'.d$	m_5
0	1	1	0	$a'.b.c.d'$	m_6
0	1	1	1	$a'.b.c.d$	m_7
1	0	0	0	$a.b'.c'.d'$	m_8
1	0	0	1	$a.b'.c'.d$	m_9
1	0	1	0	$a.b'.c.d'$	m_{10}
1	0	1	1	$a.b'.c.d$	m_{11}
1	1	0	0	$a.b.c'.d'$	m_{12}
1	1	0	1	$a.b.c'.d$	m_{13}
1	1	1	0	$a.b.c.d'$	m_{14}
1	1	1	1	$a.b.c.d$	m_{15}

Hàm Logic

- Maxterns

x	y	Maxterm	Notation
0	0	$x + y$	M_0
0	1	$x + y'$	M_1
1	0	$x' + y$	M_2
1	1	$x' + y'$	M_3

Hàm Logic

- Maxterns

x	y	z	Maxterm	Notation
0	0	0	$x + y + z$	M_0
0	0	1	$x + y + z'$	M_1
0	1	0	$x + y' + z$	M_2
0	1	1	$x + y' + z'$	M_3
1	0	0	$x' + y + z$	M_4
1	0	1	$x' + y + z'$	M_5
1	1	0	$x' + y' + z$	M_6
1	1	1	$x' + y' + z'$	M_7

Hàm Lo

- Maxterns

a	b	c	d	Maxterm	Notation
0	0	0	0	$a+b+c+d$	M_0
0	0	0	1	$a+b+c+d'$	M_1
0	0	1	0	$a+b+c'+d$	M_2
0	0	1	1	$a+b+c'+d'$	M_3
0	1	0	0	$a+b'+c+d$	M_4
0	1	0	1	$a+b'+c+d'$	M_5
0	1	1	0	$a+b'+c'+d$	M_6
0	1	1	1	$a+b'+c'+d'$	M_7
1	0	0	0	$a'+b+c+d$	M_8
1	0	0	1	$a'+b+c+d'$	M_9
1	0	1	0	$a'+b+c'+d$	M_{10}
1	0	1	1	$a'+b+c'+d'$	M_{11}
1	1	0	0	$a'+b'+c+d$	M_{12}
1	1	0	1	$a'+b'+c+d'$	M_{13}
1	1	1	0	$a'+b'+c'+d$	M_{14}
1	1	1	1	$a'+b'+c'+d'$	M_{15}

Hàm Logic

- Hàm Logic theo Minterns và Maxterns

Input Variables			Minterm		Maxterm		Complement of y	Product Term	Function Output
x	y	z					y′	y′z	F
0	0	0	m_0	$x'y'z'$	M_0	$x+y+z$	1	0	0
0	0	1	m_1	$x'y'z$	M_1	$x+y+z'$	1	1	1
0	1	0	m_2	$x'yz'$	M_2	$x+y'+z$	0	0	0
0	1	1	m_3	$x'yz$	M_3	$x+y'+z'$	0	0	0
1	0	0	m_4	$xy'z'$	M_4	$x'+y+z$	1	0	1
1	0	1	m_5	$xy'z$	M_5	$x'+y+z'$	1	1	1
1	1	0	m_6	xyz'	M_6	$x'+y'+z$	0	0	1
1	1	1	m_7	xyz	M_7	$x'+y'+z'$	0	0	1

Hàm Logic

- Hàm Logic theo Minterns và Maxterns

$$F = m_1 + m_4 + m_5 + m_6 + m_7$$
$$= \sum m(1, 4, 5, 6, 7)$$
$$= x'y'z + xy'z' + xy'z + xyz' + xyz$$

$$F = m_1 + m_4 + m_5 + m_6 + m_7$$
$$= x'y'z + xy'z' + xy'z + xyz' + xyz$$
$$= x'y'z + xy'(z' + z) + xy(z' + z)$$
$$= x'y'z + xy'.1 + xy.1$$
$$= x'y'z' + xy' + xy$$
$$= x'y'z + x(y' + y)$$
$$= x'y'z + x.1$$
$$= x'y'z + x$$
$$= (x' + x)(y'z + x)$$
$$= 1.y'z + x$$
$$= x + y'z$$

Hàm Logic

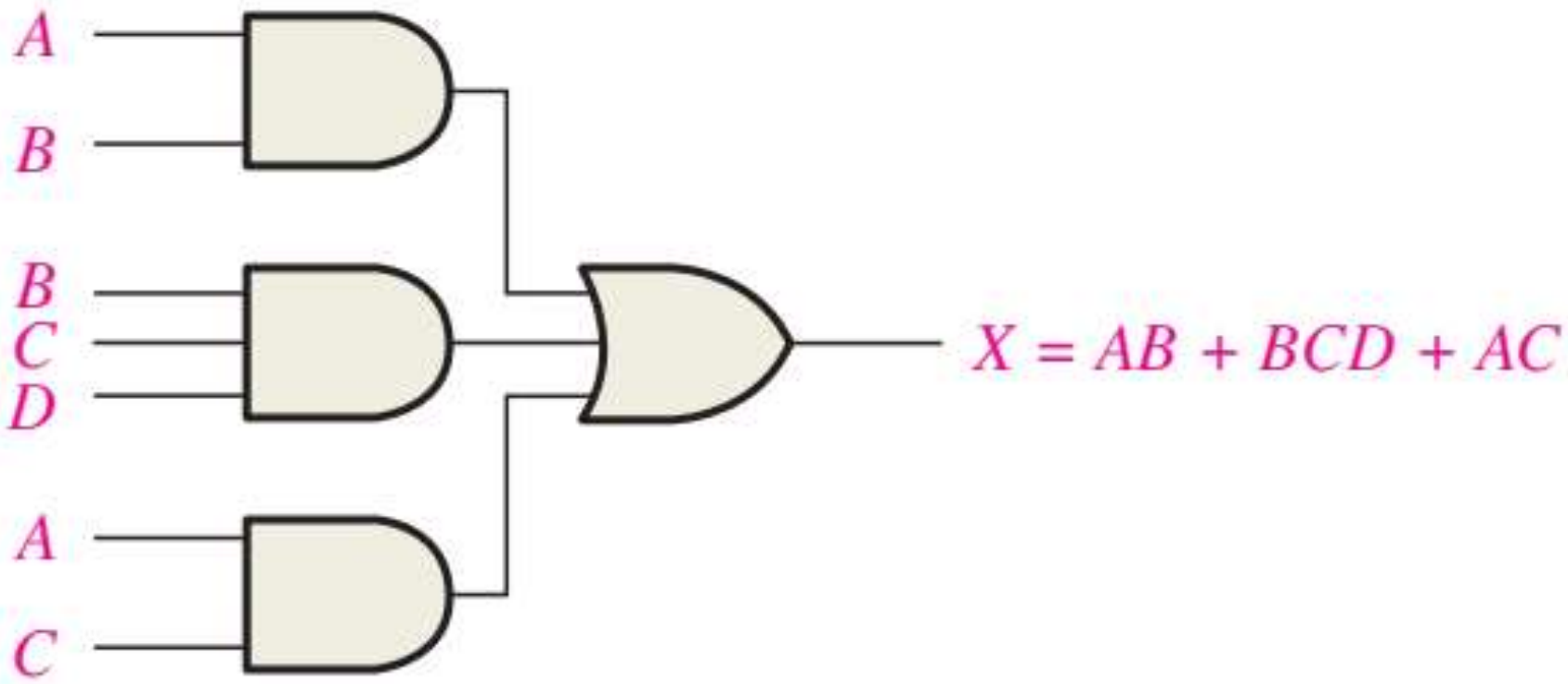
- Hàm Logic theo Minterns và Maxterns

$$F = M_0 M_2 M_3$$
$$= \prod M(0, 2, 3)$$
$$= (x + y + z)(x + y' + z)(x + y' + z')$$

$$F = M_0 M_2 M_3$$
$$= (x + y + z)(x + y' + z)(x + y' + z')$$
$$= (x + z + y)(x + z + y')(x + y' + z')$$
$$= (x + z + (y y'))(x + y' + z')$$
$$= (x + z + 0)(x + y' + z')$$
$$= x + (z(y' + z'))$$
$$= x + zy' + zz'$$
$$= x + y'z + 0$$
$$= x + y'z$$

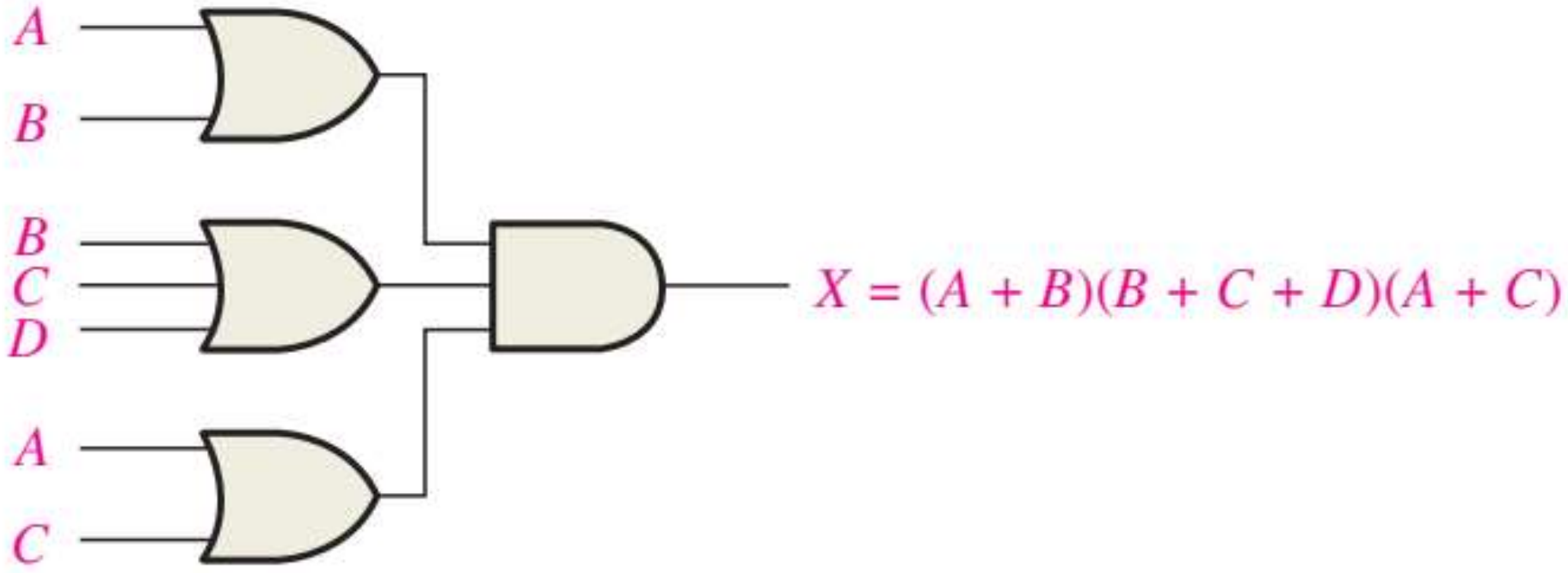
Hàm Logic

- Dạng SOP (Sum - of - Products)



Hàm Logic

- Dạng POS (Product - of - Sums)



Hàm Logic

- Chuyển biểu thức SOP sang bảng trạng thái

$$\overline{A}\,\overline{B}C + A\overline{B}\,\overline{C} + ABC$$

Inputs			Output	
A	B	C	X	Product Term
0	0	0	0	
0	0	1	1	$\overline{A}\,\overline{B}C$
0	1	0	0	
0	1	1	0	
1	0	0	1	$A\overline{B}\,\overline{C}$
1	0	1	0	
1	1	0	0	
1	1	1	1	ABC

Hàm Logic

- Chuyển biểu thức POS sang bảng trạng thái

$$(A + B + C)(A + \overline{B} + C)(A + \overline{B} + \overline{C})(\overline{A} + B + \overline{C})(\overline{A} + \overline{B} + C)$$

Inputs			Output	
A	B	C	X	Sum Term
0	0	0	0	$(A + B + C)$
0	0	1	1	
0	1	0	0	$(A + \overline{B} + C)$
0	1	1	0	$(A + \overline{B} + \overline{C})$
1	0	0	1	
1	0	1	0	$(\overline{A} + B + \overline{C})$
1	1	0	0	$(\overline{A} + \overline{B} + C)$
1	1	1	1	

Hàm Logic

- Xác định biểu thức SOP và POS từ bảng trạng thái

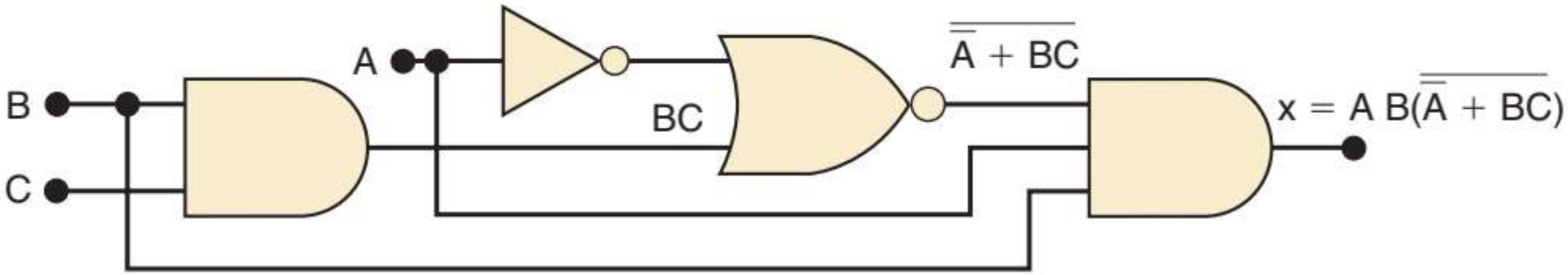
Inputs			Output
A	*B*	*C*	*X*
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1

$$X = \overline{A}BC + A\overline{B}\,\overline{C} + AB\overline{C} + ABC$$

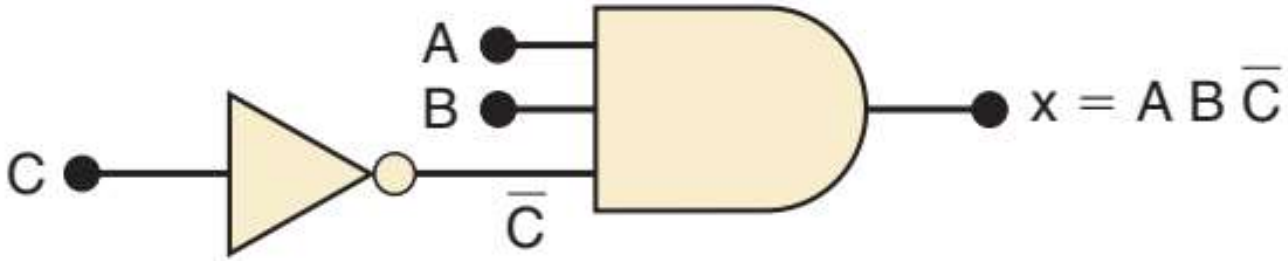
$$X = (A + B + C)(A + B + \overline{C})(A + \overline{B} + C)(\overline{A} + B + \overline{C})$$

Hàm Logic

- Tối giản biểu thức logic

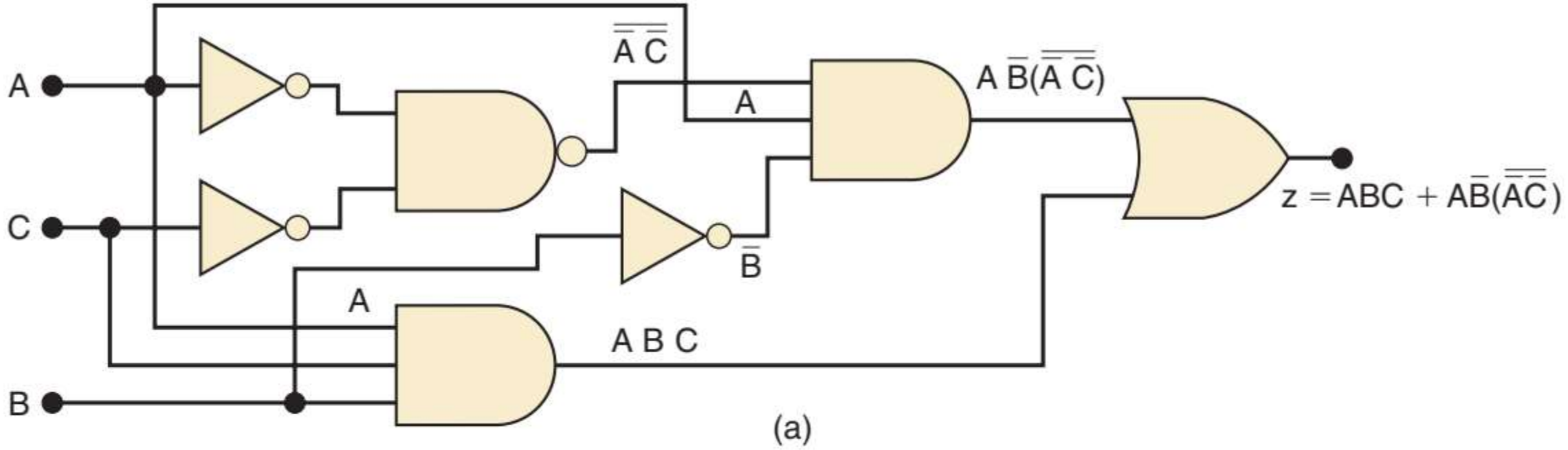


(a)

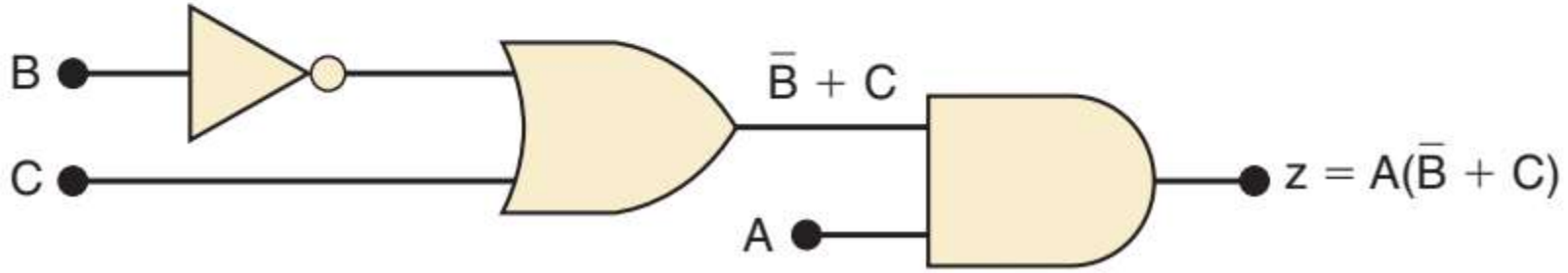


(b)

Hàm Logic


A
C
B
$\overline{\overline{A}\,\overline{C}}$
A
$A\,\overline{B}(\overline{\overline{A}\,\overline{C}})$
$\overline{B}$
A
A B C
$z = ABC + A\overline{B}(\overline{\overline{A}\,\overline{C}})$

(a)


B
C
$\overline{B} + C$
A
$z = A(\overline{B} + C)$

(b)

Bìa Karnaugh

- Bìa Karnaugh 3 biến

$AB \backslash C$	0	1
00		
01		
11		
10		

$AB \backslash C$	0	1
00	$\bar{A}\bar{B}\bar{C}$	$\bar{A}\bar{B}C$
01	$\bar{A}B\bar{C}$	$\bar{A}BC$
11	$AB\bar{C}$	ABC
10	$A\bar{B}\bar{C}$	$A\bar{B}C$

Bìa Karnaugh

- Bìa Karnaugh 4 biến

AB \ CD	00	01	11	10
00				
01				
11				
10				

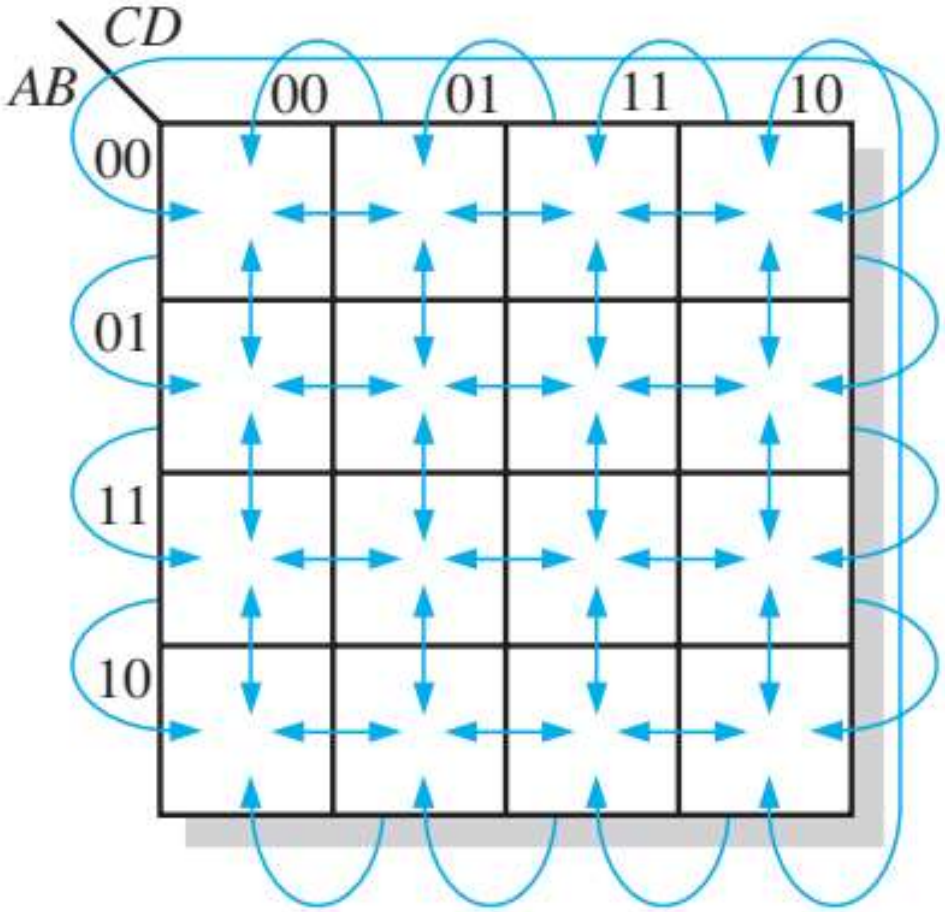
(a)

AB \ CD	00	01	11	10
00	$\bar{A}\bar{B}\bar{C}\bar{D}$	$\bar{A}\bar{B}\bar{C}D$	$\bar{A}\bar{B}CD$	$\bar{A}\bar{B}C\bar{D}$
01	$\bar{A}B\bar{C}\bar{D}$	$\bar{A}B\bar{C}D$	$\bar{A}BCD$	$\bar{A}BC\bar{D}$
11	$AB\bar{C}\bar{D}$	$AB\bar{C}D$	$ABCD$	$ABC\bar{D}$
10	$A\bar{B}\bar{C}\bar{D}$	$A\bar{B}\bar{C}D$	$A\bar{B}CD$	$A\bar{B}C\bar{D}$

(b)

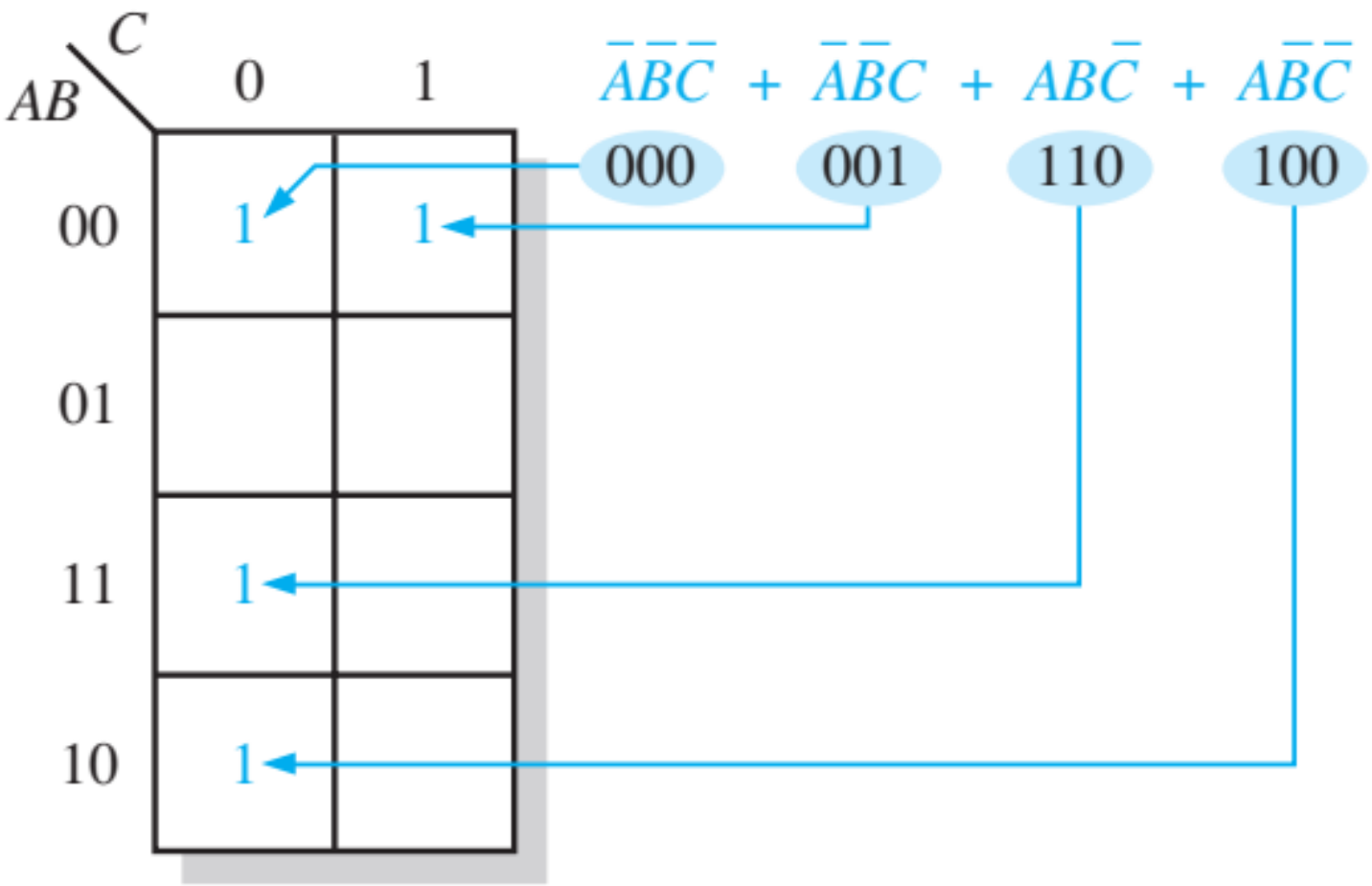
Bìa Karnaugh

- Bìa Karnaugh có tính liền kề nhau giữa các ô



Bìa Karnaugh

- Tối giản biểu thức logic SOP bằng bìa Karnaugh



Bìa Karnaugh

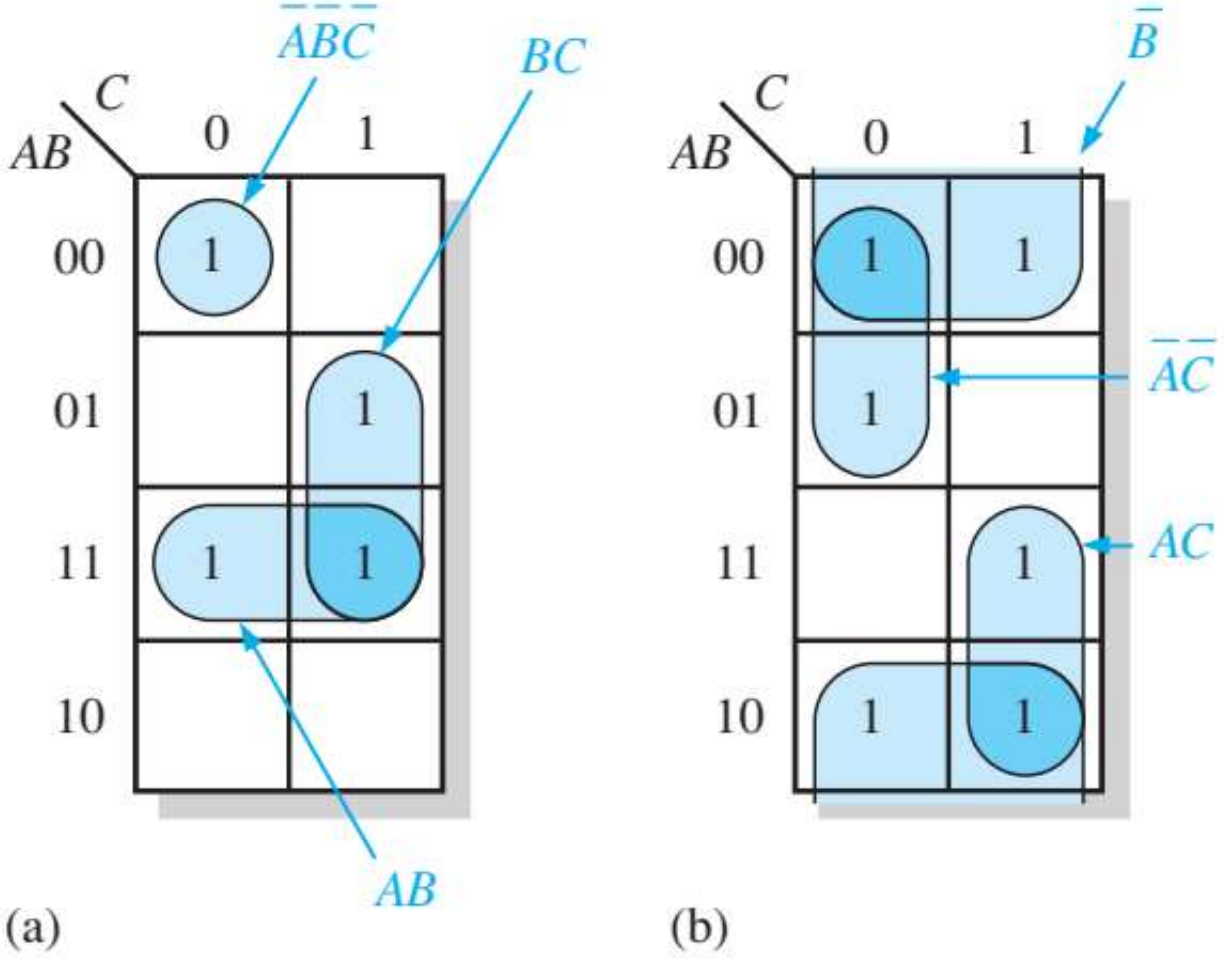
- Tối giản biểu thức logic dung bìa Karnaugh

$$\overline{A} + A\overline{B} + AB\overline{C}.$$

AB \ C	0	1
00	1	1
01	1	1
11	1	
10	1	1

Bìa Karnaugh

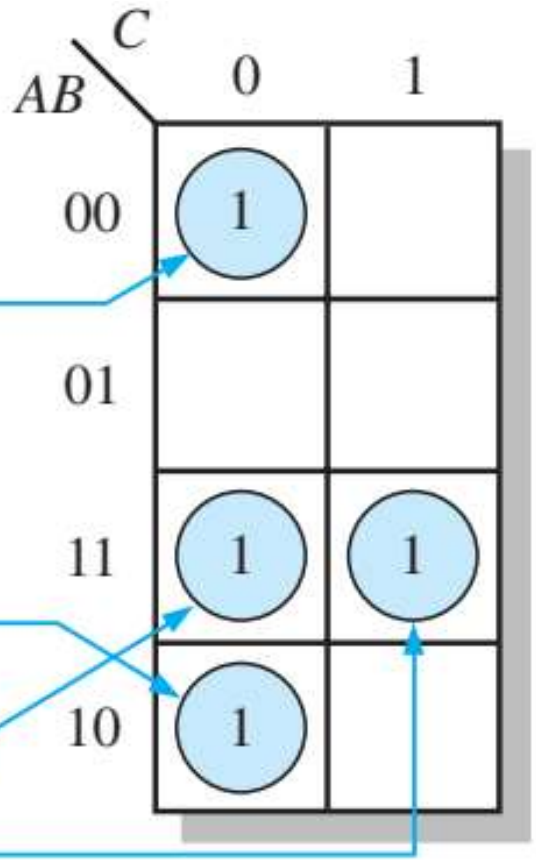
- Tối giản biểu thức logic SOP bằng bìa Karnaugh



Bìa Karnaugh

- Tối giản biểu thức logic SOP bằng bìa Karnaugh

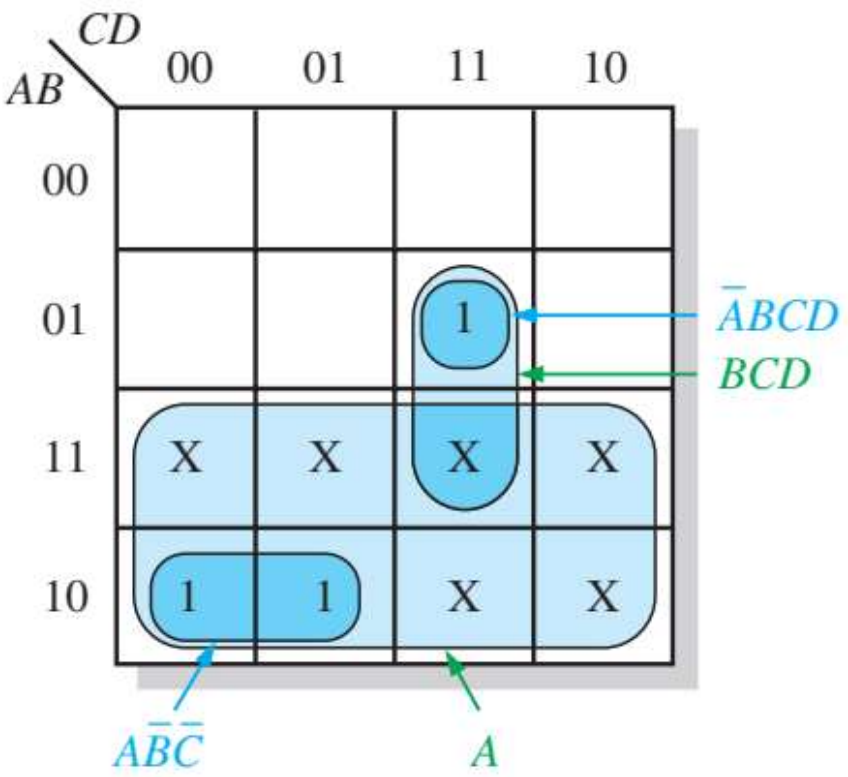
Inputs			Output
A	*B*	*C*	*X*
0	0	0	1
0	0	1	0
0	1	0	0
0	1	1	0
1	0	0	1
1	0	1	0
1	1	0	1
1	1	1	1



Inputs				Output
A	*B*	*C*	*D*	*Y*
0	0	0	0	0
0	0	0	1	0
0	0	1	0	0
0	0	1	1	0
0	1	0	0	0
0	1	0	1	0
0	1	1	0	0
0	1	1	1	1
1	0	0	0	1
1	0	0	1	1
1	0	1	0	X
1	0	1	1	X
1	1	0	0	X
1	1	0	1	X
1	1	1	0	X
1	1	1	1	X

Don't cares

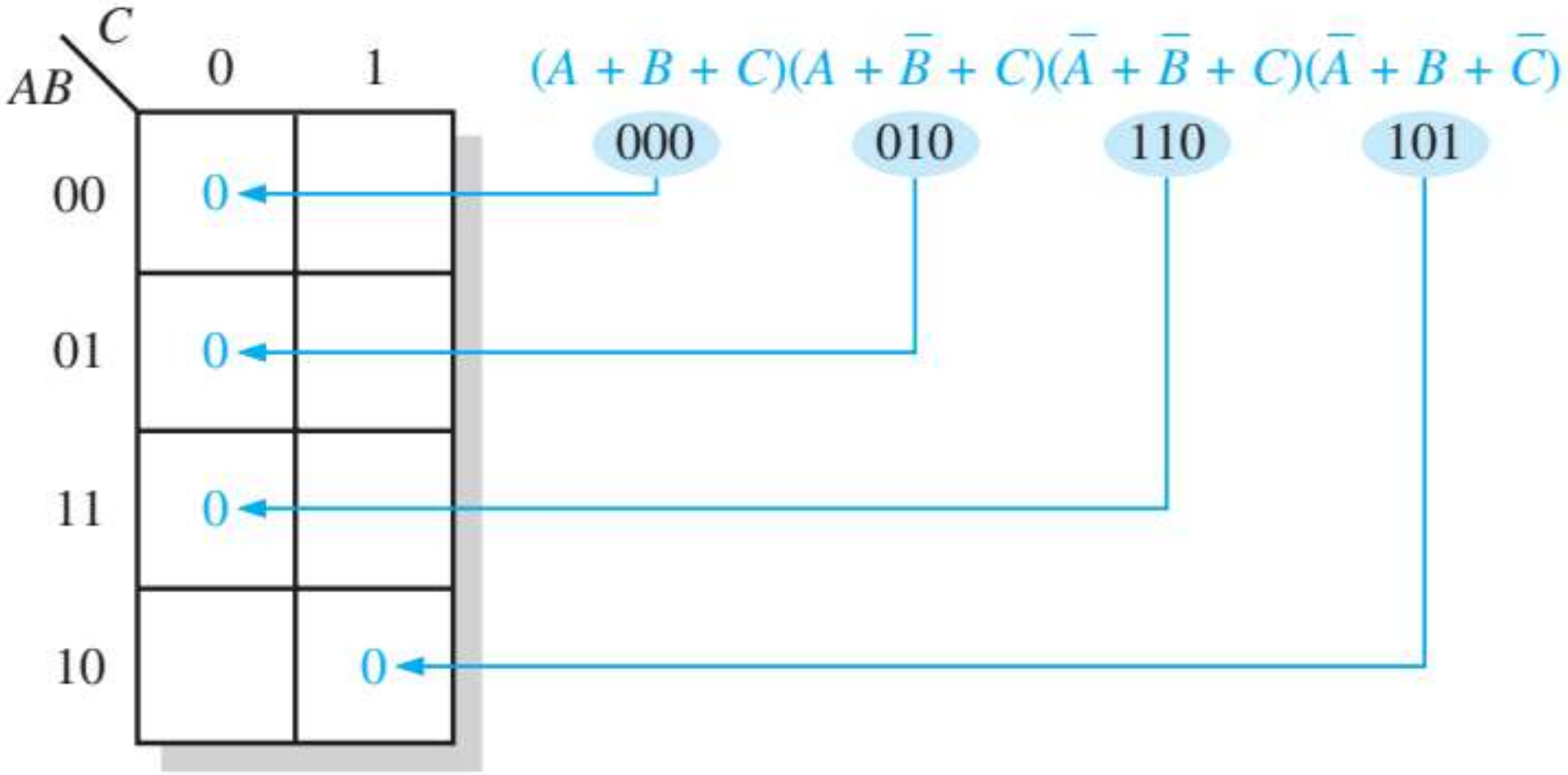
(a) Truth table



(b) Without "don't cares" $Y = A\bar{B}\bar{C} + \bar{A}BCD$
With "don't cares" $Y = A + BCD$

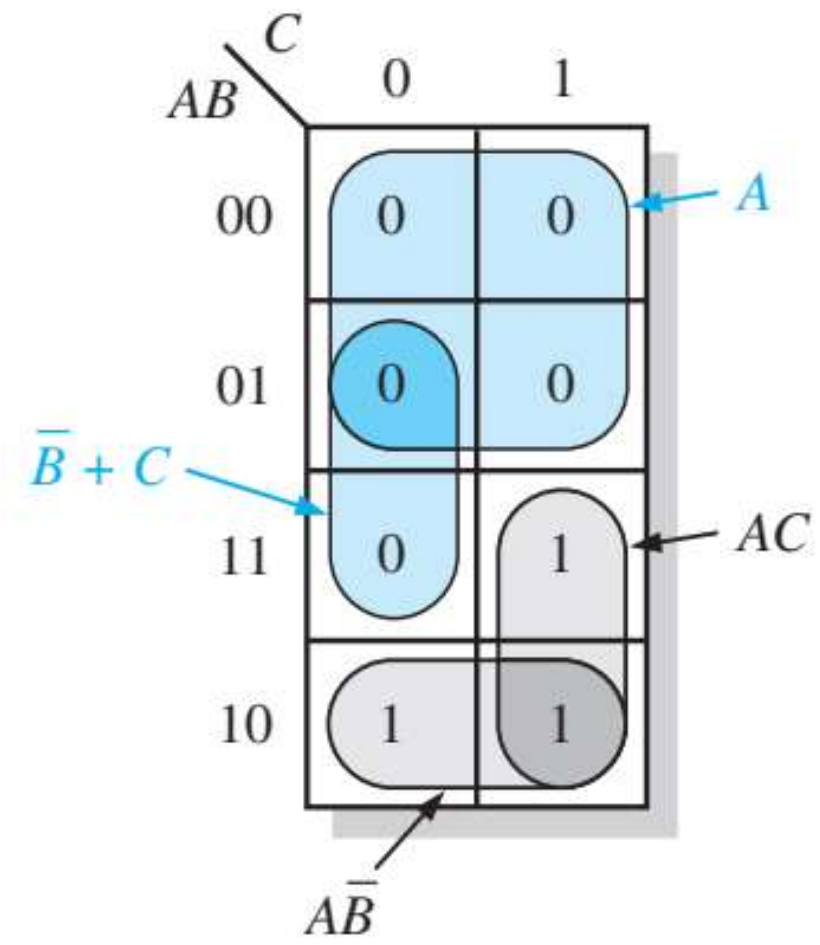
Bìa Karnaugh

- Tối giản biểu thức logic POS bằng bìa Karnaugh



Bìa Karnaugh

- Tối giản biểu thức logic POS bằng bìa Karnaugh



Bìa Karnaugh

- Tối giản biểu thức logic SOP bằng bìa Karnaugh

$$F(A, B, C) = \sum m(0, 1, 4, 5, 7)$$

Inputs			Minterm	Output
A	B	C		F
0	0	0	m_0	1
0	0	1	m_1	1
0	1	0	m_2	0
0	1	1	m_3	0
1	0	0	m_4	1
1	0	1	m_5	1
1	1	0	m_6	0
1	1	1	m_7	1

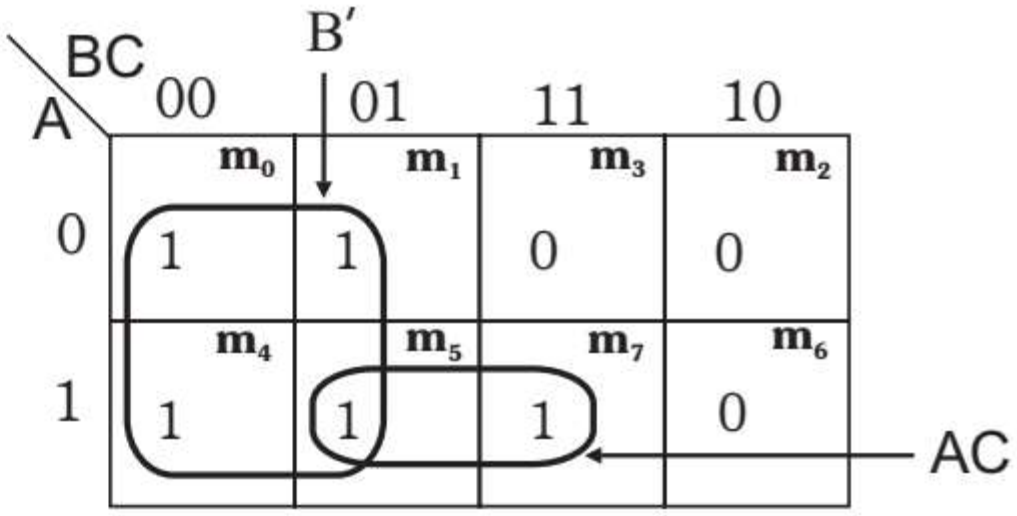


Fig. 1: K-map for F.

Bìa Karnaugh

- Tối giản biểu thức logic POS bằng bìa Karnaugh

$F_1(A, B, C, D) = \prod M(1, 3, 5, 7, 13, 15)$

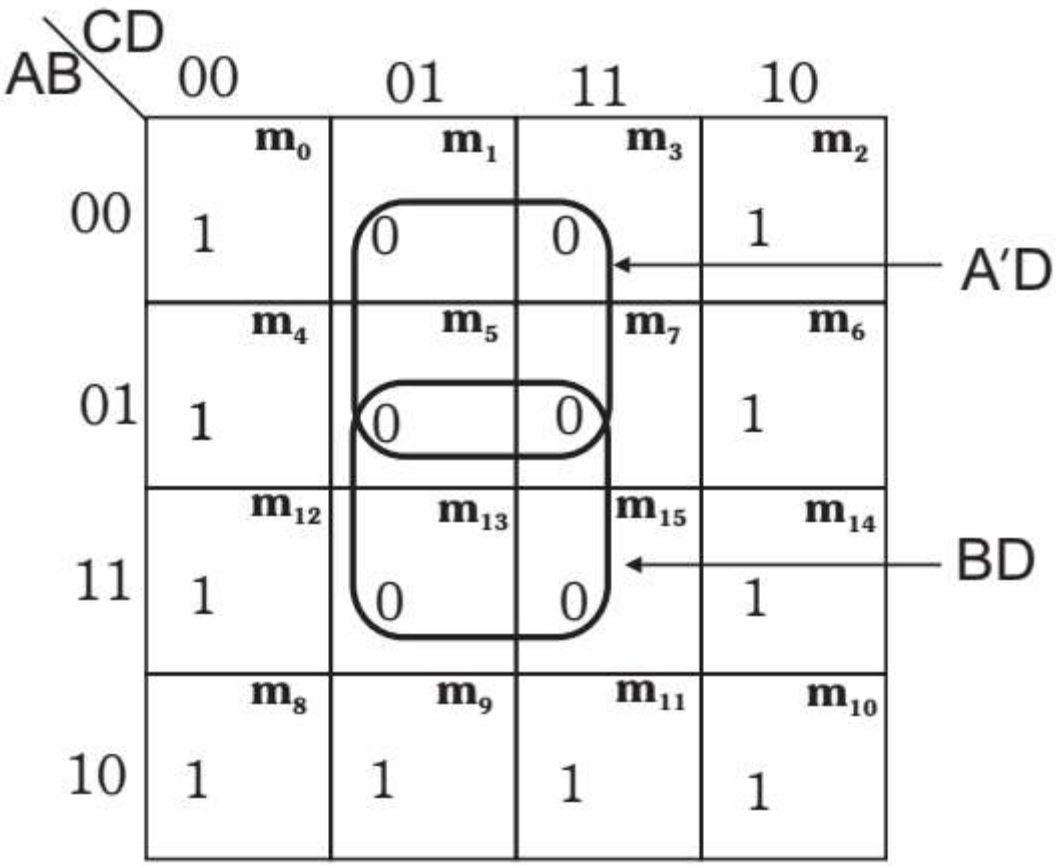


Fig. 1: K-map for F_1.

Mạch Logic

- Thiết kế mạch logic

- Thiết lập bảng sự thật.
- Viết **AND** /OR cho các ngõ ra = 1/0.
- Viết biểu thức dạng SOP /POS.
- Tối giản biểu thức logic.
- Hoàn thành mạch logic từ biểu thức tối giản.

Mạch Logic

- Thiết kế mạch logic

Design a logic circuit with three inputs, A, B, and C. Output to be HIGH only when a majority inputs are HIGH.

Step 1. Set up the truth table.

A	B	C	x
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1

Mạch Logic

- Thiết kế mạch logic

Step 2. Write the AND term for each case where the output is a 1.

A	B	C	x	
0	0	0	0	
0	0	1	0	
0	1	0	0	
0	1	1	1	$\rightarrow \overline{A}BC$
1	0	0	0	
1	0	1	1	$\rightarrow A\overline{B}C$
1	1	0	1	$\rightarrow AB\overline{C}$
1	1	1	1	$\rightarrow ABC$

Mạch Logic

- Thiết kế mạch logic

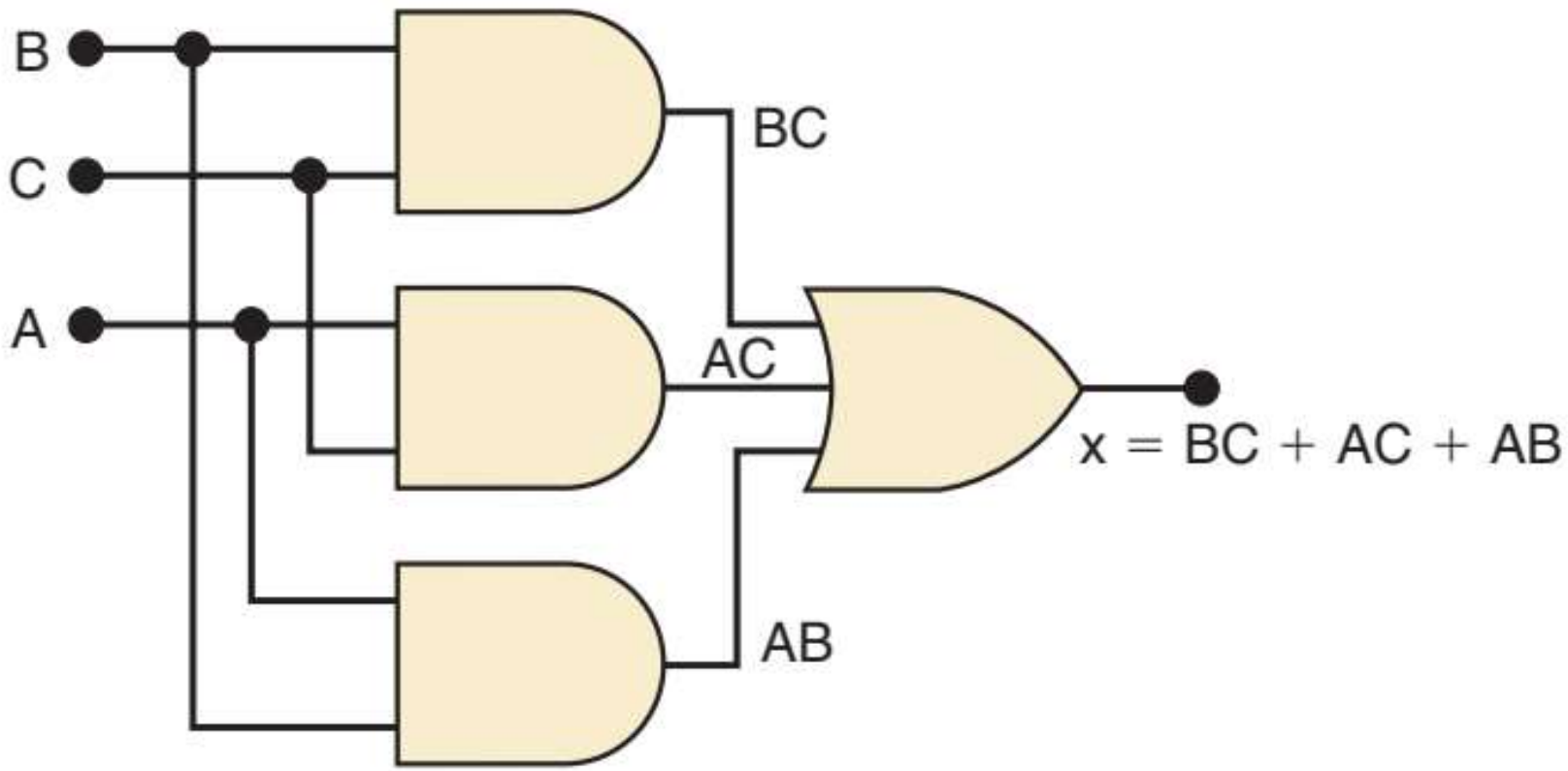
Step 3. Write the sum-of-products expression for the output.

$$x = \overline{A}BC + A\overline{B}C + AB\overline{C} + ABC$$

Step 4. Simplify the output expression.

$$x = BC + AC + AB$$

Mạch Logic

- Thiết kế mạch logic

Step 5. Implement the circuit for the final expression.